

# Monopoly Rents, International Trade, and Structural Change

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September 26, 2026

## Abstract

The divergence between sectoral employment and value-added shares presents a longstanding puzzle in theories of structural change. This paper develops a Schumpeterian model in which technological innovation and international trade in intermediate varieties provide a mechanism for this divergence. In a closed economy, value-added and employment shares coincide because sectoral profits scale with sectoral employment. Opening the economy breaks this equivalence. When a country holds the global technological lead in a larger share of a sector's intermediate varieties, its firms earn monopoly profits from both domestic and foreign demand, increasing domestic value added relative to employment. Conversely, when the country imports a larger share of the relevant varieties, the associated monopoly rents accrue abroad, reducing domestic value added relative to employment. The model therefore links the gap between value-added and employment shares to relative technological leadership, intermediate-good trade, and foreign demand. In particular, this gap widens with intermediate-good exports and higher foreign prices and narrows with intermediate-good imports. These predictions are consistent with sector-level evidence from 42 countries over 2000–2014, where the gap between value-added and employment shares varies systematically with intermediate-goods trade and the profit share of value added.

*Keywords:* Schumpeterian growth, structural change, innovation, monopoly rents, international trade, value-added and employment shares

*JEL Classification:* F12, O14, O31, O33, O41

# 1 Introduction

A central concept in development economics is structural change, defined as the reallocation of economic resources across sectors with different productivity levels. Three measures are commonly used to characterize this process: sectoral employment shares, sectoral value-added shares, and sectoral final-consumption-expenditure shares. Although these measures are often treated as closely related, they can differ substantially. [Herrendorf et al. \(2014\)](#) document systematic differences among them, while [Kuznets \(1966\)](#) shows that during the early stages of U.S. development, the employment share of services increased substantially while its value-added share remained relatively constant. [Buera and Kaboski \(2009\)](#) identify this divergence as a puzzle for theories of structural transformation across countries: value-added and employment shares both measure the composition of domestic production, yet standard production theory provides no mechanism for why they should systematically diverge. This paper develops a new theory in which international trade in intermediate varieties and the resulting location of monopoly profits generate the wedge between sectoral value-added and employment shares.

The distinction between the three measures of structural change matters because they capture different margins of economic activity. Employment and value added are production measures, whereas final-consumption expenditure records where the value of goods and services is ultimately spent. International trade can therefore generate a gap between production and consumption measures: domestic production can be sold abroad, while domestic consumption can be supplied by foreign producers. Indeed, [Herrendorf et al. \(2014\)](#) attribute part of the divergence between production and consumption measures to international trade and differences in measurement conventions. But this explanation does not resolve a different and more basic empirical puzzle: in the data, why do two measures of domestic production, sectoral employment and sectoral value added, systematically diverge? This paper shows that trade in intermediate varieties provides a mechanism for this divergence because it changes the location of monopoly profits relative to employment.

Using sectoral data from the World Input-Output Database, I document three facts that motivate the theoretical analysis. First, value-added and employment shares differ systematically across sectors. Manufacturing has a value-added share above its employment share in most observations, agriculture exhibits the opposite pattern, and services are more evenly distributed around the one-to-one relationship. Second, the gap between value-added and employment shares varies systematically with intermediate-goods trade. It is larger where intermediate-good exports are greater and smaller where intermediate-good imports are greater, conditional on the other trade margins and fixed effects. Third, the same export and import margins are related to the profit share of value added in the corresponding directions. These patterns connect the value-added–employment share gap to intermediate-goods trade and to the location of profits, suggesting that the divergence between the two production measures reflects where the returns to production accrue rather than where employment is located.

Therefore, I develop a Schumpeterian model of structural change to provide a mechanism for these patterns. Technological innovation drives sectoral productivity growth, while three final sectors, agriculture, manufacturing, and services, produce distinct consumption goods competitively using labor and a continuum of sector-specific intermediate varieties. Intermediate varieties are produced under monopolistic competition, protected by patents, and traded internationally. The model combines the Schumpeterian innovation framework of [Aghion and Howitt \(2009\)](#) with a two-country, three-sector setting in which intermediate varieties are supplied to the world market by the country holding the most

advanced version of each variety. This structure allows the market for a successful innovation to extend beyond the country in which the innovator and associated employment are located, separating the location of employment from the location of monopoly profits. On the demand side, I adopt the non-homothetic constant elasticity of substitution (CES) preferences of [Comin et al. \(2021\)](#), which allow income elasticities to differ across sectors.

The closed-economy benchmark establishes why the distinction between employment and value added is not generated by innovation alone. In the absence of international trade, the value-added share of every sector equals its employment share. Sectoral profits are proportional to employment in the sector, so sectoral value added, consisting of wages and profits, is proportional to sectoral employment. The two shares therefore coincide exactly. Technological innovation can generate productivity differences and structural change, but it does not by itself create a wedge between the two production measures ([Matsuyama, 2019](#)).

International trade in intermediate varieties breaks this equivalence by expanding the market over which the technology leader can earn monopoly rents. When a country is the global technology leader in a sector, its firms supply patented intermediate varieties to final-good producers both at home and abroad. Foreign demand therefore expands the market for these varieties, increasing the scale of monopoly sales and the profits accruing to the firms that hold the leading technologies. Because these profits accrue to the owners of the technologies, foreign demand raises the value added attributed to the technology-leading country relative to the employment associated with domestic final-good production. The country's sectoral value-added share therefore rises relative to its employment share, generating a positive value-added–employment share gap. When the country instead imports the relevant intermediate varieties, the corresponding monopoly rents accrue to foreign producers. Domestic employment can still be associated with the production of final goods that use these imported varieties, while the associated monopoly rents are captured abroad. Domestic value added therefore excludes these foreign rents, generating a negative value-added–employment share gap. The model thus links the direction of the value-added–employment share gap to a country's position in international trade in intermediate varieties and, through that position, to the location of the associated monopoly rents.

The model also helps explain why the value-added–employment share gap differs systematically across sectors and countries. A sector in which domestic firms do not hold the leading technologies cannot generate monopoly rents from foreign demand for those technologies, while reliance on foreign intermediate technologies directs such rents to producers abroad. The model therefore predicts that sectors in which domestic firms relatively lead and export innovative intermediate varieties should tend to have value-added shares above their employment shares, whereas sectors that rely on foreign intermediate technologies should tend to have value-added shares below their employment shares. This distinction helps explain the patterns observed in the data: agriculture, where countries rarely occupy the leading technological position, exhibits a negative gap in almost all observations, while manufacturing, where technological leadership and intermediate-good exports are more prevalent, exhibits a positive gap in most observations.<sup>1</sup>

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<sup>1</sup>The mechanism does not imply a uniform value-added–employment share gap across development levels. Rather, the gap depends on the sectors in which a country controls valuable intermediate inputs and supplies them to foreign markets. Mining provides an example: the discovery or control of scarce natural resources can generate rents from foreign demand without a proportional increase in domestic employment, much as technological leadership can.

**Related Literature.** The paper thus connects three strands of the structural-change literature that are usually considered separately: the measurement of structural change, the role of international trade in shaping production and consumption shares, and innovation-driven changes in sectoral productivity and rents. The central result is that employment and value-added shares need not coincide even when both are measured from domestic production. Once innovative intermediate varieties are traded internationally, the location of monopoly rents can differ from the location of employment, creating a systematic wedge between the two measures. The model generates relationships linking this wedge to intermediate-good exports, intermediate-good imports, technological leadership, and foreign demand.

The literature on structural change traces to [Lewis \(1954\)](#), [Chenery \(1960\)](#), [Kuznets \(1966\)](#), [Baumol \(1967\)](#), and [Harris and Todaro \(1970\)](#), who established surplus labor, industrialization, income-inequality dynamics, the cost-disease phenomenon, and rural-urban migration as central mechanisms of development. Subsequent work formalized several of these mechanisms and showed how differences in productivity growth, preferences, and relative prices shape the reallocation of resources across sectors. [Ngai and Pissarides \(2007\)](#) formalize the Baumol price effect, whereby differential sectoral productivity growth changes relative prices and thereby affects sectoral demand. [Matsuyama \(2009\)](#) and [Uy et al. \(2013\)](#) incorporate international trade into theories of structural change, showing how trade openness and comparative advantage affect sectoral specialization, production, and employment patterns.<sup>2</sup>

A related literature has treated the divergence between value-added and employment shares as a puzzle for theories of structural change. Standard models imply that the two shares should coincide ([Herrendorf et al., 2014](#); [Buera and Kaboski, 2009](#)), yet the data show systematic differences between them. More recently, [Sáenz \(2022\)](#) shows that time-varying capital intensities can generate such a divergence in South Korea. [Matsuyama \(2019\)](#) shows how non-homothetic demand shapes sectoral employment and trade in a monopolistically competitive economy with free entry. With free entry, however, profits are competed away, so value added equals the wage bill and value-added and employment shares move together. The mechanism developed here instead introduces patent-protected monopoly rents in intermediate-goods production. International trade then determines whether those rents accrue to domestic or foreign producers, allowing value-added and employment shares to diverge within country-sector observations, and linking the direction of the gap to a country's position in international trade in intermediate varieties.

The contribution is therefore not to reinterpret trade as a source of structural change in general, but to identify a specific channel through which trade can separate two measures of domestic production. The model shows that trade in final goods can separate production from consumption, as emphasized in the existing literature ([Uy et al., 2013](#); [Herrendorf et al., 2014](#)), while trade in innovative intermediate varieties can separate value added from employment by changing where monopoly rents are earned. This distinction provides a unified mechanism connecting technological leadership, intermediate-goods trade, sectoral profits, and the value-added–employment share gap. The empirical evidence presented in the paper motivates this mechanism by showing that the gap varies systematically with intermediate-good exports and imports and that the same trade margins are related to the profit share of value added.

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<sup>2</sup>In addition, [Duarte and Restuccia \(2020\)](#), [Rodrik \(2016\)](#), [Sposi \(2019\)](#), [Avoumatsodo and Leunga Noukwe \(2024\)](#), and [Swiecki \(2017\)](#) examine how the forces underlying structural change vary across stages of development.

**Outline.** The remainder of the paper proceeds as follows. Section 2 presents the motivating empirical evidence. Section 3 develops the closed-economy model and shows the value-added–employment equivalence. Section 4 introduces international trade, derives the value-added–employment wedge, and studies its comparative statics. Section 5 concludes.

## 2 Motivating Evidence

This section documents three facts. First, the value-added–employment gap differs systematically across sectors. Second, the gap is positively related to intermediate-good exports and negatively related to intermediate-good imports. Third, the same pattern appears in the profit share of value added, linking the observed gap to the profit channel formalized in the model. These facts motivate the model, which explains why trade generates a systematic wedge between value-added and employment shares.

### 2.1 Data

I use the 2016 release of the World Input-Output Database (WIOD) (Timmer, Dietzenbacher, Los, Stehrer and de Vries, 2015), which provides annual data on value added, employment, gross output, labor and capital compensation, and bilateral trade flows for 43 countries over 2000–2014.<sup>3</sup> I aggregate the 56 WIOD industries (ISIC Revision 4) into three broad economic sectors: agriculture (ISIC sections A01–A03), manufacturing (ISIC sections B–E39),<sup>4</sup> and services (ISIC sections F–U98, spanning construction, wholesale and retail trade, transportation, accommodation and food services, information and communication, finance and insurance, real estate, professional and administrative services, public administration, education, health, and other services). All sectoral measures are constructed from this same industry aggregation. The resulting country-sector-year panel is balanced, with 42 countries, three sectors, and 15 years, yielding 1,890 observations.<sup>5</sup>

For each country, sector, and year, I measure the value-added–employment wedge as the log ratio of the sector’s value-added share to its employment share. I then examine how this wedge varies with the composition of international trade. For each country, sector, and year, I construct intermediate-good exports and imports, final-consumption-good exports and imports, and investment-good exports and imports. Each trade measure is normalized by sectoral value added. This normalization expresses trade relative to the economic size of the sector, rather than its absolute volume, so that a large and a small sector can be compared on the same scale. It also measures the intensity of foreign demand for (or foreign supply to) the sector relative to the value it generates domestically. Intermediate-good exports capture

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<sup>3</sup>The countries are Australia, Austria, Belgium, Bulgaria, Brazil, Canada, Switzerland, China, Cyprus, the Czech Republic, Germany, Denmark, Spain, Estonia, Finland, France, the United Kingdom, Greece, Croatia, Hungary, Indonesia, India, Ireland, Italy, Japan, South Korea, Lithuania, Luxembourg, Latvia, Mexico, Malta, the Netherlands, Norway, Poland, Portugal, Romania, Russia, Slovakia, Slovenia, Sweden, Turkey, Taiwan, and the United States.

<sup>4</sup>The “manufacturing” sector combines mining and quarrying (ISIC B), manufacturing proper (ISIC C10–C33), and electricity, gas, water, and waste services (ISIC D35, E36–E39) into a single broad industrial category, matching how the sector is defined in the theoretical model of Section 3.

<sup>5</sup>Luxembourg is dropped from the analysis because its manufacturing sector’s intermediate-export and -import intensities are pinned at the sample’s 99th-percentile winsorization ceiling in 8 and 7 of its 15 sample years, respectively—the highest hit-rate of any country-sector series in the sample (compare Belgium’s manufacturing sector, the next most affected, at 6 and 5 years). This pattern is consistent with Luxembourg’s well-documented role as a hub for holding companies and phantom foreign investment (Damgaard et al., 2019), reflecting a trade-to-GDP ratio far outside the range of any other sample country.

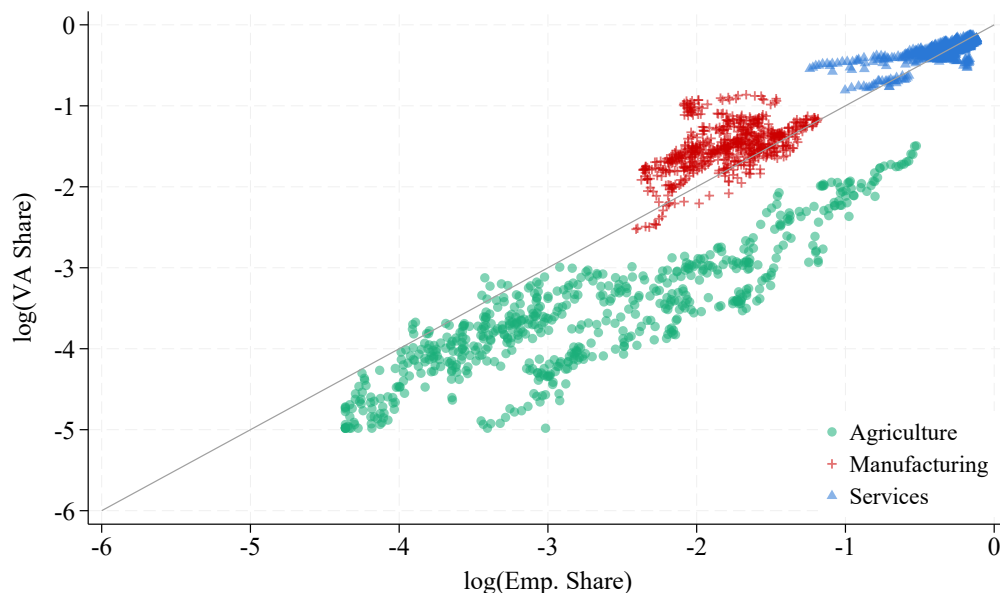
domestic output sold abroad for intermediate use, while intermediate-good imports capture foreign output entering domestic intermediate use. Foreign final-demand flows are separated into consumption and investment according to their use.

To examine the profit channel directly, I use the share of value added accruing as capital compensation. Capital compensation is the residual between value added and labor compensation in the WIOD national accounts and provides the closest national-accounts counterpart to the portion of value added not accruing to labor. It therefore provides a direct empirical link between the observed value-added–employment wedge and the profit mechanism developed in the model. Because labor compensation for self-employed workers is imputed in WIOD, capital compensation can be negative when the imputed labor cost exceeds measured value added. These observations are concentrated in agriculture and in Taiwan’s service sector. The logarithm of the capital-compensation share is therefore defined only for observations with a positive share.

## 2.2 Facts

I now turn to what these variables show: a persistent, sector-asymmetric gap between value-added and employment shares that tracks a country’s net position in intermediate-goods trade and, through the same channel, its profit share of value added.

**Value-Added vis-à-vis Employment Shares.** Figure 1 plots the log value-added share,  $\log(va_{jkt})$ , against the log employment share,  $\log(emp_{jkt})$ , for each country  $j$ , sector  $k$ , and year  $t$ . On this log-log scale, the vertical distance from the 45-degree line is  $\log(va_{jkt}/emp_{jkt})$  at every point.



**Figure 1: Value-Added Share vs. Employment Share**

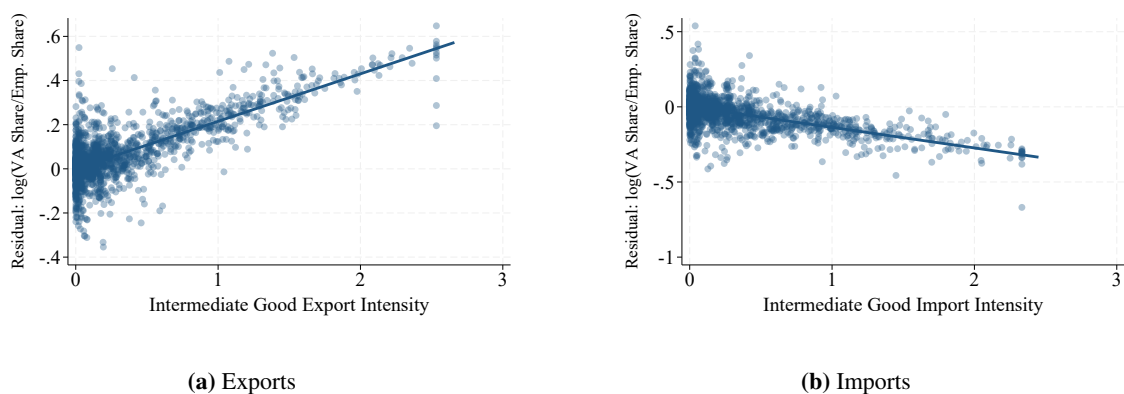
*Notes:* This figure shows the log value-added (VA) share plotted against the log employment share for every country-sector-year observation, pooled across all 42 countries. Diagonal: 45-degree benchmark (equal shares); vertical distance from it equals  $\log(\text{VA Share}/\text{Emp. Share})$ . Data: WIOD 2016, 42 countries, 3 broad sectors, 2000–2014.

The figure first confirms a well-documented fact in the structural-change literature: value-added and

employment shares do not generally coincide (Buera and Kaboski, 2009; Herrendorf et al., 2014; Sáenz, 2022). The divergence is substantial, but it is also strongly sectoral. Manufacturing, shown by red plus signs, has value-added shares above employment shares in 86.5% of observations. Agriculture, shown by filled green circles, exhibits the opposite pattern, with value-added shares below employment shares in 98.1% of observations. Services, shown by blue triangles, is nearly evenly split, with 48.1% of observations above the 45-degree line. These differences are consistent with the sectors' different exposure to tradable, monopolizable varieties (Comin et al., 2025). Manufacturing is the sector in which global monopoly rights over traded intermediate varieties are most concentrated, whereas agriculture is more globally competitive and labor-intensive, with less scope for monopoly rents from traded intermediates. The service sector is more heterogeneous: much of its activity, including retail, personal services, and other largely non-traded activities, lies outside the international monopoly-rights mechanism, while a smaller traded component, including finance, ICT, and business services, may participate in it.

**Intermediate-Goods Trade and the Value-Added–Employment Wedge.** A country's position in intermediate-goods trade is closely related to the value-added–employment gap: the gap is larger where intermediate-good exports are higher and smaller where intermediate-good imports are higher. These relationships remain after accounting for the other trade margins, country-sector fixed effects, and year fixed effects.

Figure 2 illustrates these relationships using the component of the log value-added–employment ratio that is orthogonal to the other trade margins and fixed effects. The broad dispersion of observations around the fitted relationships indicates that the patterns are not driven by a few country-sector-year observations. The corresponding partial-residual plots for investment-good and final-consumption trade are reported in Figures B.1 and B.2 in the Appendix B. They show less systematic relationships with the value-added–employment share gap than those for intermediate-goods trade.



**Figure 2:** Intermediate-Goods Trade and the Value-Added–Employment Gap

*Notes:* This figure shows partial-residual plots, following Larsen and McCleary (1972), of  $\log(\text{VA Share}/\text{Emp. Share})$  against intermediate exports intensity (panel a) and intermediate imports intensity (panel b), pooled across all three sectors. Data: WIOD 2016, 42 countries, 3 broad sectors, 2000–2014.

Formally, I estimate the following equation:

$$\log(va_{jkt}/emp_{jkt}) = \rho_0 + \rho_1 Exp_{jkt} + \rho_2 Imp_{jkt} + \rho_3 X_{jkt} + \phi_j + \phi_k + \phi_t + u_{jkt}, \quad (2.1)$$

where  $va_{jkt}$  and  $emp_{jkt}$  denote the value-added and employment shares of country  $j$  in sector  $k$  and year  $t$ , respectively.  $Exp_{jkt}$  and  $Imp_{jkt}$  denote intermediate-good exports and imports, each normalized by sectoral value added.  $X_{jkt}$  contains the remaining trade margins, including final-consumption and investment-good exports and imports. These controls account for other forms of international trade that may be correlated with intermediate-goods trade and the value-added–employment gap. Their inclusion therefore isolates the relationship between intermediate-goods trade and the gap from broader differences in the composition of international trade. The terms  $\phi_j$ ,  $\phi_k$ , and  $\phi_t$  denote country, sector, and year fixed effects, respectively, and  $u_{jkt}$  is the error term. The coefficients  $\rho_1$  and  $\rho_2$  capture the relationships between intermediate-goods trade and the value-added–employment gap, conditional on the other trade margins and fixed effects.

**Table 1:** Value-Added–Employment Log Ratio: Regression Results

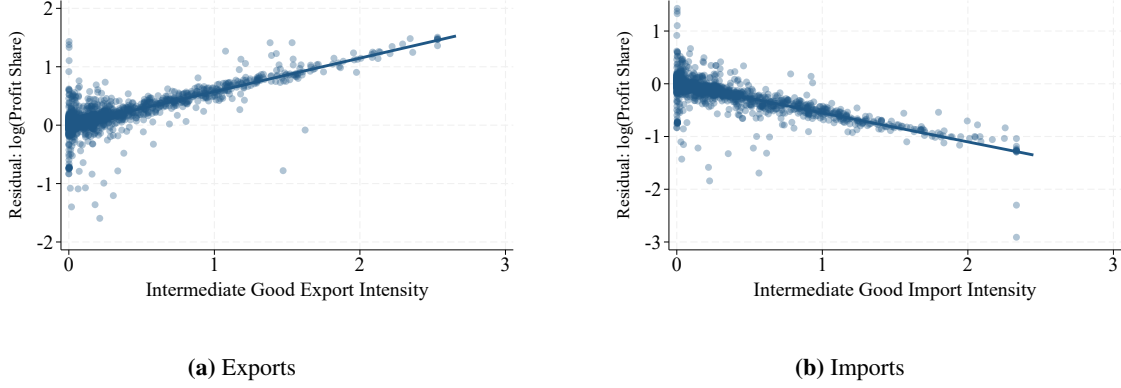
|                            | (1)                  | (2)                   | (3)                  | (4)                   | (5)                  | (6)                   |
|----------------------------|----------------------|-----------------------|----------------------|-----------------------|----------------------|-----------------------|
| Intermediate Goods Export  | 0.440***<br>(0.124)  | 0.186**<br>(0.0750)   | 0.494**<br>(0.185)   | 0.180**<br>(0.0751)   | 0.410*<br>(0.214)    | 0.215**<br>(0.0984)   |
| Intermediate Goods Import  | -0.735***<br>(0.157) | -0.187***<br>(0.0568) | -0.555***<br>(0.202) | -0.208***<br>(0.0472) | -0.775***<br>(0.192) | -0.137***<br>(0.0383) |
| Investment Goods Export    |                      |                       | -0.403<br>(0.544)    | 0.342**<br>(0.134)    | -0.446<br>(0.496)    | 0.320**<br>(0.150)    |
| Investment Goods Import    |                      |                       | -0.901**<br>(0.372)  | -0.0794<br>(0.158)    | -0.947**<br>(0.464)  | -0.0472<br>(0.173)    |
| Consumption-Goods Export   |                      |                       |                      |                       | 0.702***<br>(0.153)  | -0.148<br>(0.103)     |
| Consumption Goods Import   |                      |                       |                      |                       | 0.0894<br>(0.0994)   | -0.167**<br>(0.0732)  |
| Observations               | 1,890                | 1,890                 | 1,890                | 1,890                 | 1,890                | 1,890                 |
| Country FE                 | Yes                  | No                    | Yes                  | No                    | Yes                  | No                    |
| Year FE                    | Yes                  | Yes                   | Yes                  | Yes                   | Yes                  | Yes                   |
| Sector FE                  | Yes                  | No                    | Yes                  | No                    | Yes                  | No                    |
| Country $\times$ Sector FE | No                   | Yes                   | No                   | Yes                   | No                   | Yes                   |
| Adj. R-squared             | 0.770                | 0.978                 | 0.778                | 0.978                 | 0.792                | 0.978                 |

*Notes:* This table presents panel regression estimates of Equation (2.1) with the log ratio  $\log(va_{jkt}/emp_{jkt})$  as the dependent variable. Data: WIOD 2016, 42 countries, 3 broad sectors, 2000–2014.  $N = 1,890$ . Standard errors clustered by country in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 1 reports estimates of Equation (2.1) using the high-dimensional fixed-effects estimator of Correia (2017), with standard errors clustered by country. The table builds the specification in two dimensions. Odd-numbered columns include separate country and sector fixed effects, while even-numbered columns replace these with country  $\times$  sector fixed effects as a robustness check. The latter absorb all time-invariant heterogeneity specific to each country-sector pair. The trade controls are added in three cumulative blocks: intermediate trade alone in columns 1–2, investment trade in columns 3–4, and consumption trade in columns 5–6.

The results show a negative and statistically significant relationship between intermediate-good imports and the value-added–employment share gap in every specification. Intermediate-good exports are also positively and significantly related to the gap throughout. The coefficients on investment-good trade and consumption-good trade change considerably across specifications, and are occasionally significant

without a stable sign across the two fixed-effects levels. Overall, the intermediate-good margins exhibit the clearest and most consistent relationship with the value-added–employment share gap.



**Figure 3:** Intermediate Trade and the Profit Share

*Notes:* This figure shows partial-residual plots of the profit share of value added against intermediate export intensity (panel a) and intermediate import intensity (panel b), pooled across all three sectors. Data: WIOD 2016, 42 countries, 3 broad sectors, 2000–2014.

**Intermediate-Goods Trade and the Profit Share of Value Added.** Intermediate-goods exports and imports are also related to the profit share of value added in the same direction as the value-added–employment gap: exports are positively related to the profit share, while imports are negatively related to it. Figure 3 uses the same partial-residual approach as the preceding analysis, restricting the sample to observations with a positive capital-compensation share ( $N \approx 1,757$ ). The dispersion around the fitted relationships is broad rather than concentrated in a small number of observations. The corresponding partial-residual plots for investment-good and final-consumption trade are reported in Figures B.3 and B.4 in the Appendix B and show less systematic relationships.

To quantify this relationship, I estimate Equation (2.2) as follows, with the profit share of value added as the dependent variable:

$$\log(\Pi_{jkt}/VA_{jkt}) = \rho_0 + \rho_1 Exp_{jkt} + \rho_2 Imp_{jkt} + \rho_3 X_{jkt} + \phi_j + \phi_k + \phi_t + u_{jkt}, \quad (2.2)$$

where  $\Pi_{jkt}$  denotes total profits and  $VA_{jkt}$  denotes nominal value added in country  $j$ , sector  $k$ , and period  $t$ . The remaining variables and fixed effects are defined as in Equation (2.1).

Table 2 reports the corresponding estimates, using the same specifications and column structure as Table 1. The profit-share results show a pattern similar to that for the value-added–employment share gap. Intermediate-good exports are positively related to the profit share of value added, whereas intermediate-good imports are negatively related to it, with both signs preserved across all specifications. These relationships also remain statistically distinguishable from zero throughout. By contrast, the investment-good coefficients lose statistical significance once country  $\times$  sector fixed effects are introduced, while the consumption-good coefficients change substantially across the fixed-effect specifications. The intermediate-good margins therefore provide the most consistent evidence of a relationship between international trade and the profit share of value added.

Taken together, the evidence shows that the gap between value-added and employment shares varies

**Table 2:** Profit Share (Log): Regression Results

|                            | (1)                  | (2)                  | (3)                 | (4)                  | (5)                  | (6)                  |
|----------------------------|----------------------|----------------------|---------------------|----------------------|----------------------|----------------------|
| Intermediate Goods Export  | 0.361***<br>(0.132)  | 0.554***<br>(0.179)  | 0.548***<br>(0.160) | 0.553***<br>(0.200)  | 0.513***<br>(0.157)  | 0.574**<br>(0.215)   |
| Intermediate Goods Import  | -0.557***<br>(0.155) | -0.530***<br>(0.157) | -0.460**<br>(0.206) | -0.542***<br>(0.148) | -0.808***<br>(0.179) | -0.551***<br>(0.168) |
| Investment Goods Export    |                      |                      | -1.096**<br>(0.428) | 0.0467<br>(0.459)    | -1.141***<br>(0.421) | 0.0237<br>(0.436)    |
| Investment Goods Import    |                      |                      | -0.427<br>(0.387)   | 0.0875<br>(0.156)    | -0.482<br>(0.501)    | 0.270<br>(0.208)     |
| Consumption-Goods Export   |                      |                      |                     |                      | 0.859***<br>(0.211)  | 0.148<br>(0.187)     |
| Consumption Goods Import   |                      |                      |                     |                      | 0.137<br>(0.119)     | -0.285**<br>(0.119)  |
| Observations               | 1,758                | 1,757                | 1,758               | 1,757                | 1,758                | 1,757                |
| Country FE                 | Yes                  | No                   | Yes                 | No                   | Yes                  | No                   |
| Year FE                    | Yes                  | Yes                  | Yes                 | Yes                  | Yes                  | Yes                  |
| Sector FE                  | Yes                  | No                   | Yes                 | No                   | Yes                  | No                   |
| Country $\times$ Sector FE | No                   | Yes                  | No                  | Yes                  | No                   | Yes                  |
| Adj. R-squared             | 0.354                | 0.790                | 0.369               | 0.790                | 0.403                | 0.791                |

*Notes:* This table reports panel regression estimates of Equation (2.2), with the log profit share as the dependent variable. Data: WIOD 2016, 42 countries, 3 broad sectors, 2000–2014.  $N \approx 1,757$ . Standard errors clustered by country are reported in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

systematically with intermediate-goods trade and that the same trade margins are related to the profit share of value added. The next section develops a theory that rationalizes these patterns through the interaction between innovation-driven monopoly power and international trade in intermediate varieties.

### 3 The Closed Economy Model

The theoretical model builds on the Schumpeterian growth framework of [Aghion and Howitt \(2009\)](#), which is extended to an economy with three final sectors, agriculture, manufacturing, and services. Each sector produces a distinct consumption good competitively using labor and a continuum of sector-specific intermediate varieties. Each variety is supplied by an incumbent producer that holds a patent over its current technology and can invest in research to raise its productivity. Successful innovation extends the producer's patent-protected monopoly over the variety. Households value the three final goods according to the non-homothetic CES preferences of [Comin et al. \(2021\)](#). They finance firms' innovation expenditures out of current income and receive the resulting profits in return. Time is discrete, and country  $j$  has a mass  $L_{jt}$  of individuals in each period. Sectoral productivity growth therefore arises from the innovation decisions of domestic firms.

#### 3.1 Goods Production Sectors

**Final Goods Production.** Each final good  $k$ , consumed by households, is produced competitively using labor and a unit interval of intermediate varieties  $v$ , according to the Cobb-Douglas production

function:

$$Y_{jkt} = L_{jkt}^{1-\alpha} \int_0^1 A_{jkt}(\nu)^{1-\alpha} x_{jkt}(\nu)^\alpha d\nu, \quad k = a, m, s; \quad (3.1)$$

where  $0 < \alpha < 1$ ,  $Y_{jkt}$  denotes output of final good  $k$  in country  $j$  at time  $t$ ,  $A_{jkt}(\nu)$  denotes the productivity of intermediate variety  $\nu$  used in sector  $k$ , and  $x_{jkt}(\nu)$  denotes the quantity of the latest version of variety  $\nu$  used in producing final good  $k$  in country  $j$  at time  $t$ . Employment in the production of final good  $k$  is denoted by  $L_{jkt}$ , with total employment satisfying:

$$\sum_{k=a,m,s} L_{jkt} = L_{jt}, \quad \forall j \in J. \quad (3.2)$$

Because final-good production is competitive, the representative firm takes the output price  $P_{jkt}$  and intermediate input prices  $p_{jkt}(\nu)$  as given. It chooses labor  $L_{jkt}$  and the quantity of each intermediate variety  $x_{jkt}(\nu)$  to maximize profit:

$$\max_{\{L_{jkt}, [x_{jkt}(\nu)]_{\nu \in [0,1]}\}} P_{jkt} L_{jkt}^{1-\alpha} \int_0^1 A_{jkt}(\nu)^{1-\alpha} x_{jkt}(\nu)^\alpha d\nu - \int_0^1 p_{jkt}(\nu) x_{jkt}(\nu) d\nu - w_{jt} L_{jkt} \quad (3.3)$$

where  $w_{jt}$  denotes the wage rate in country  $j$  at time  $t$ .

**Intermediate Goods Production.** Each intermediate variety  $\nu$  used in sector  $k$  is supplied by a patent monopolist that holds the patent for as long as it remains the most advanced producer of that variety. Producing one unit of intermediate variety  $\nu$  requires one unit of final good  $k$ , so the marginal cost of producing the intermediate is  $P_{jkt}$ . Innovation is assumed to be drastic, so the patent monopolist faces no effective competition from less advanced producers. The monopolist producing variety  $\nu$  for sector  $k$  in country  $j$  therefore chooses its quantity to maximize profit:

$$\max_{\{x_{jkt}(\nu)\}} \pi_{jkt}(\nu) = p_{jkt}(\nu) x_{jkt}(\nu) - P_{jkt} x_{jkt}(\nu) \quad (3.4)$$

$$\text{s.t.} \quad p_{jkt}(\nu) = f^{-1} [x_{jkt}(\nu)]$$

where  $f$  denotes the final-good producer's demand function for intermediate variety  $\nu$ .

### 3.2 Innovation and Productivity Growth

Productivity growth arises from innovation by the incumbent producer of each intermediate variety. In each period, the incumbent can invest in research and development (R&D) to raise the productivity of the variety it supplies: this is the own-innovation margin of Schumpeterian growth theory, in which the current producer of a variety improves it directly, as distinct from the replacement margin, in which a rival displaces the current producer outright. A successful innovation increases productivity by a factor  $\gamma_{jk} > 1$ , while an unsuccessful attempt leaves productivity unchanged; production therefore continues in either case, at whichever productivity level results. The closed-economy model considers this own-innovation margin; the open-economy model additionally allows a foreign entrepreneur to compete for

the same variety, generating the replacement margin.<sup>6</sup>

Let  $\mu_{jkt}(\mathbf{v})$  denote the probability of successful innovation in variety  $\mathbf{v}$  of sector  $k$  in country  $j$ . Productivity evolves according to

$$A_{jkt+1}(\mathbf{v}) = \begin{cases} \gamma_{jk} A_{jkt}(\mathbf{v}) & \text{with probability } \mu_{jkt}(\mathbf{v}) \\ A_{jkt}(\mathbf{v}) & \text{with probability } 1 - \mu_{jkt}(\mathbf{v}) \end{cases}$$

The probability of success depends on R&D spending according to

$$\lambda_j \frac{P_{jkt} Z_{jkt}(\mathbf{v})}{\gamma_{jk} A_{jkt}(\mathbf{v})} = F[\mu_{jkt}(\mathbf{v})], \quad F' > 0, F'' > 0, F(0) = 0, \quad (3.5)$$

where  $\lambda_j$  captures the extent to which national policies and institutions support innovation, and  $P_{jkt} Z_{jkt}(\mathbf{v})$  denotes research-and-development (R&D) expenditure on variety  $\mathbf{v}$ . The denominator  $\gamma_{jk} A_{jkt}(\mathbf{v})$  measures the productivity level targeted by the innovation. Thus, for a given R&D expenditure, innovation becomes more difficult as the target productivity level rises. Since  $F' > 0$ , Equation (3.5) is invertible, so choosing R&D spending  $Z_{jkt}(\mathbf{v})$  is equivalent to choosing the innovation probability  $\mu_{jkt}(\mathbf{v})$ .

The incumbent chooses R&D spending to maximize the expected discounted profit from the variety net of the R&D cost:

$$\max_{\{Z_{jkt}(\mathbf{v})\}} \beta \left[ \mu_{jkt}(\mathbf{v}) \pi_{jkt+1}(\mathbf{v}) \Big|_{\gamma_{jk} A_{jkt}(\mathbf{v})} + (1 - \mu_{jkt}(\mathbf{v})) \pi_{jkt+1}(\mathbf{v}) \Big|_{A_{jkt}(\mathbf{v})} \right] - P_{jkt} Z_{jkt}(\mathbf{v}) \quad (3.6)$$

where  $\pi_{jkt+1}(\mathbf{v}) \Big|_{\gamma_{jk} A_{jkt}(\mathbf{v})}$  and  $\pi_{jkt+1}(\mathbf{v}) \Big|_{A_{jkt}(\mathbf{v})}$  denote Equation (3.19) evaluated at next period's productivity under success and failure, respectively. Because a failed innovation leaves the incumbent's productivity unchanged, the associated monopoly profit continues to be earned. The cost of failure is therefore the R&D expenditure itself, while a successful innovation additionally raises the future profit from the variety.

### 3.3 Household

A representative household in country  $j$  supplies  $L_{jt}$  units of labor inelastically at wage  $w_{jt}$ . Labor is perfectly mobile across the three final sectors. The household has no access to saving or borrowing and therefore consumes its current income each period. The household's problem can thus be solved period by period. Following [Comin et al. \(2021\)](#), preferences over the three sectoral composite goods are represented by a non-homothetic CES aggregator. Let  $C_{jkt}$  denote consumption of the composite good produced in sector  $k$ , where  $k = a, m, s$ , and define the consumption index  $C_{jt}$  implicitly by

$$\sum_{k=a,m,s} \delta_k^{1/\sigma} \left( \frac{C_{jkt}}{C_{jt}^{\varepsilon_k}} \right)^{\frac{\sigma-1}{\sigma}} = 1, \quad (3.7)$$

<sup>6</sup>The distinction between own innovation and replacement follows the Schumpeterian growth framework of [Aghion and Howitt \(2009\)](#). In the closed economy, no domestic rival is specified for a variety. In the open economy, foreign entrepreneurs can compete for the same variety; the resulting innovation race is described in Appendix A.6. The same setup applies to each of the three sectors.

where  $\delta_k$  are constant sectoral weights satisfying  $\sum_{k=a,m,s} \delta_k = 1$ ,  $\sigma < 1$  is the elasticity of substitution between sectoral composite goods, and  $\varepsilon_k$  is the income elasticity of demand for sector  $k$ . Thus, sectors with  $\varepsilon_k > 1$  have expenditure shares that rise with income, while  $\varepsilon_k < 1$  implies the opposite. The consumption index  $C_{jt}$  is defined implicitly by Equation (3.7) and serves only as a theoretical representation of preferences; sectoral expenditure  $P_{jkt}C_{jkt}$  is the corresponding observable object.<sup>7</sup>

When  $\varepsilon_k = 1$  for all  $k$ , Equation (3.7) reduces to the standard homothetic CES aggregator:

$$C_{jt} = \left( \sum_{k \in \{a,m,s\}} \delta_k^{1/\sigma} C_{jkt}^{1-\frac{1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad \text{if } \varepsilon_k = 1 \quad \forall k = a, m, s; \quad (3.8)$$

so that homothetic preferences are the special case in which all sectoral income elasticities equal one.

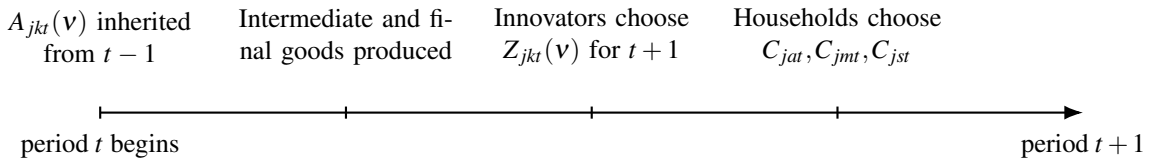
In each period, the household chooses sectoral consumption to maximize the consumption index:

$$\begin{aligned} & \max_{\{C_{jat}, C_{jmt}, C_{jst}\}} \log C_{jt} \\ \text{s.t.} \quad & \sum_{k \in \{a,m,s\}} P_{jkt} (C_{jkt} + Z_{jkt}) \leq w_{jt} L_{jt} + \sum_{k=a,m,s} \Pi_{jkt} \end{aligned} \quad (3.9)$$

where  $\Pi_{jkt} := \int_0^1 \pi_{jkt}(v) dv$  is total profit earned in sector  $k$  of country  $j$ . R&D spending enters the budget constraint because the household is the residual claimant of firm profits and finances innovation out of current income. The resulting profits are realized in the following period, as specified in Equation (3.6). The household problem is therefore equivalent to choosing sectoral consumption to minimize expenditure subject to the non-homothetic CES aggregator.

### 3.4 Equilibrium

The timing of the model is as follows. At the beginning of period  $t$ , each intermediate variety inherits its productivity from the previous period. Final-good firms then produce using the available intermediate varieties, while intermediate producers supply these varieties at their monopoly prices. Given current production and prices, innovators choose R&D spending for the next period. Households then allocate their current income across the three final goods.<sup>8</sup>



**Figure 4:** Timing of the model within period  $t$ .

<sup>7</sup>The hand-to-mouth structure keeps the household side simple and isolates the innovation and trade mechanisms. It also avoids introducing an intertemporal saving margin that is not needed for the sectoral allocation analyzed here. The non-homothetic CES specification allows expenditure shares to vary systematically with income while retaining a constant elasticity of substitution across sectoral goods.

<sup>8</sup>The timing convention implies that R&D expenditure is incurred in period  $t$ , while the productivity outcome and associated profits are realized in period  $t + 1$ .

**Definition 1** (Closed-economy equilibrium). *A closed-economy equilibrium for country  $j$  consists of sequences of wages and final-good prices,*

$$\mathbf{p}_j = \{w_{jt}, P_{jat}, P_{jmt}, P_{jst}\}_{t=0}^{\infty}; \quad (3.10)$$

*household consumption allocations,*

$$\mathbf{c}_j = \{C_{jat}, C_{jmt}, C_{jst}\}_{t=0}^{\infty}; \quad (3.11)$$

*labor and intermediate-input allocations for final-good firms,*

$$\mathbf{f}_j = \{L_{jkt}, x_{jkt}(\mathbf{v})\}_{t=0; k=a,m,s}^{\infty}; \quad (3.12)$$

*and R&D and intermediate-good price decisions,*

$$\mathbf{z}_j = \{Z_{jat}, Z_{jmt}, Z_{jst}\}_{t=0}^{\infty} \quad \text{and} \quad \mathbf{p}_j^x = \{p_{jat}(\mathbf{v}), p_{jmt}(\mathbf{v}), p_{jst}(\mathbf{v})\}_{t=0; \mathbf{v} \in [0,1]}^{\infty}. \quad (3.13)$$

*These sequences constitute an equilibrium if, given prices, household consumption solves Equation (3.9), final-good firms solve their profit-maximization problem, and intermediate producers solve Equations (3.4) and (3.6). Labor and goods markets must also clear.*

**Market Clearing Conditions.** The factor and goods market clearing conditions in each period are:

$$L_{jt} = L_{jat} + L_{jmt} + L_{jst}. \quad (3.14)$$

$$Y_{jkt} = C_{jkt} + \int_0^1 [x_{jkt}(\mathbf{v}) + Z_{jkt}(\mathbf{v})] d\mathbf{v}. \quad (3.15)$$

### 3.5 Firms' Optimization

The final-good firm's first-order conditions in sector  $k$  are:

$$\begin{cases} p_{jkt}(\mathbf{v}) = \alpha P_{jkt} x_{jkt}(\mathbf{v})^{\alpha-1} A_{jkt}(\mathbf{v})^{1-\alpha} L_{jkt}^{1-\alpha} & \forall \mathbf{v} \in [0, 1] \\ w_{jt} = (1 - \alpha) P_{jkt} L_{jkt}^{-\alpha} \int_0^1 A_{jkt}(\mathbf{v})^{1-\alpha} x_{jkt}(\mathbf{v})^{\alpha} d\mathbf{v} \end{cases}$$

The first condition gives the firm's demand for intermediate variety  $\mathbf{v}$ :

$$x_{jkt}(\mathbf{v}) = \alpha^{\frac{1}{1-\alpha}} \left( \frac{p_{jkt}(\mathbf{v})}{P_{jkt}} \right)^{-\frac{1}{1-\alpha}} A_{jkt}(\mathbf{v}) L_{jkt}, \quad \forall \mathbf{v} \in [0, 1] \quad \text{and } k = a, m, s. \quad (3.16)$$

Substituting Equation (3.16) into the intermediate producer's problem in Equation (3.4) gives the equilibrium quantity of variety  $\mathbf{v}$ :

$$x_{jkt}(\mathbf{v}) = \alpha^{\frac{2}{1-\alpha}} A_{jkt}(\mathbf{v}) L_{jkt}, \quad (3.17)$$

and its price:

$$p_{jkt}(v) = \alpha^{-1} P_{jkt}. \quad (3.18)$$

Thus, the intermediate monopolist charges a constant markup over its marginal cost. Its equilibrium profit is

$$\pi_{jkt}(v) = \pi P_{jkt} A_{jkt}(v) L_{jkt}, \quad (3.19)$$

where  $\pi := (1 - \alpha)\alpha^{\frac{1+\alpha}{1-\alpha}}$ . Profits therefore increase with the productivity of the variety, the size of the sector, as measured by employment, and the price of the final good.

Substituting Equation (3.19) into the two branches of Equation (3.6) gives:

$$\pi_{jkt+1}(v) \Big|_{\gamma_{jk} A_{jkt}(v)} = \pi P_{jkt+1} \gamma_{jk} A_{jkt}(v) L_{jkt+1} \quad (\text{success}),$$

$$\pi_{jkt+1}(v) \Big|_{A_{jkt}(v)} = \pi P_{jkt+1} A_{jkt}(v) L_{jkt+1} \quad (\text{failure}).$$

The expected payoff therefore consists of the profit the incumbent earns regardless of the innovation outcome and the additional profit generated by a successful innovation. Only the latter depends on the innovation decision. The innovator's problem in Equation (3.6) consequently reduces to:

$$\max_{\{\mu_{jkt}(v)\}} \left[ \beta \pi (\gamma_{jk} - 1) \mu_{jkt}(v) P_{jkt+1} L_{jkt+1} - \lambda_j^{-1} P_{jkt} F(\mu_{jkt}(v)) \gamma_{jk} \right] A_{jkt}(v) \quad (3.20)$$

which yields a common innovation probability across varieties within sector  $k$ :

$$\mu_{jkt} = F'^{-1} \left[ \beta \pi \lambda_j \frac{P_{jkt+1}}{P_{jkt}} L_{jkt+1} \frac{\gamma_{jk} - 1}{\gamma_{jk}} \right], \quad \forall v \in [0, 1] \text{ and } \forall k = a, m, s. \quad (3.21)$$

For the special case  $F(\mu_{jkt}(v)) = \frac{1}{2} \mu_{jkt}(v)^2$ , the innovation probability becomes:

$$\mu_{jkt} = \beta \pi \lambda_j \frac{P_{jkt+1}}{P_{jkt}} L_{jkt+1} \frac{\gamma_{jk} - 1}{\gamma_{jk}}, \quad \forall k = a, m, s. \quad (3.22)$$

The innovation probability increases with the size of the sector and with the growth rate of its output price. A larger sector provides a larger market for a successful innovation, while a higher expected output price raises the value of the resulting monopoly profit. The factor  $(\gamma_{jk} - 1)/\gamma_{jk}$  captures the fact that innovation provides an incremental gain over the profit the incumbent already earns: failure leaves the incumbent in place and does not eliminate its existing profit.

**Remark 3.1** (Markup invariance). *The equilibrium markup implied by Equation (3.18),  $p_{jkt}(v)/P_{jkt} = \alpha^{-1}$ , depends only on the intermediate-good production technology. The open-economy pricing rule derived in Section 4, Equation (4.3), has the same form. Opening the economy to trade therefore does not change the markup charged by the intermediate monopolist. The effect of trade on the value-added–employment share gap instead operates through the scale of demand for the monopolist's varieties, including demand from foreign final-good producers when the country is the technology leader.*

### 3.6 Aggregate Behavior

Define the productivity of sector  $k$  in country  $j$  at time  $t$  as the average productivity across its continuum of intermediate varieties:

$$A_{jkt} := \int_0^1 A_{jkt}(v) dv. \quad (3.23)$$

Using the expression for  $\mu_{jkt}$  in Equation (3.22), and after simplification,<sup>9</sup> the productivity growth rate of sector  $k$  is:

$$g_{A_{jkt}} = \beta \pi \lambda_j \frac{P_{jkt+1}}{P_{jkt}} \frac{(\gamma_{jk} - 1)^2}{\gamma_{jk}} L_{jkt+1}. \quad (3.24)$$

Because all varieties innovate independently with the same probability  $\mu_{jkt}$ , the law of large numbers implies that the fraction of varieties that successfully innovate equals  $\mu_{jkt}$ . Aggregate sectoral productivity growth is therefore deterministic, even though innovation at the variety level is stochastic (Grossman and Helpman, 1991).

Equation (3.24) shows that productivity growth increases with sectoral employment and with the growth of the sectoral output price. A larger sector provides a larger market for intermediate varieties, increasing monopoly profits and therefore the return to successful innovation. Sectoral demand and productivity growth are thus interdependent: changes in market size affect innovation incentives, while productivity growth in turn affects the evolution of sectoral prices and quantities. This interaction links the demand-driven mechanism emphasized by Engel's law to the supply-side mechanism of productivity growth (Matsuyama, 2019).

Substituting Equation (3.17) into Equation (3.1) gives equilibrium output of final good  $k$ :

$$Y_{jkt} = \alpha^{\frac{2\alpha}{1-\alpha}} A_{jkt} L_{jkt}, \quad (3.25)$$

and the final-sector first-order conditions imply a common wage:

$$w_{jt} = \omega P_{jkt} A_{jkt}, \quad (3.26)$$

where  $\omega := (1 - \alpha) \alpha^{\frac{2\alpha}{1-\alpha}}$ . Since the wage is common across sectors, Equation (3.26) implies that relative sectoral prices are inversely related to relative productivity:

$$\frac{P_{kt}}{P_{mt}} = \frac{A_{mt}}{A_{kt}}. \quad (3.27)$$

Thus, slower productivity growth in a sector leads to a higher relative price over time.

Let  $VA_{jkt}$  denote value added in sector  $k$  and its associated intermediate production in country  $j$  at

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<sup>9</sup>See Appendix A.3.

time  $t$ . Then<sup>10</sup>

$$\begin{aligned} VA_{jkt} &= P_{jkt}Y_{jkt} - P_{jkt} \int_0^1 x_{jkt}(v) dv \\ &= \zeta P_{jkt}A_{jkt}L_{jkt} = (1 + \alpha)w_{jt}L_{jkt}, \end{aligned} \quad (3.28)$$

where  $\zeta := (1 - \alpha^2)\alpha^{\frac{2\alpha}{1-\alpha}}$ . The last equality uses Equation (3.26) and  $\zeta/\omega = 1 + \alpha$  to express value added solely in terms of the common wage and sectoral employment.

Summing Equation (3.28) across sectors gives aggregate GDP:

$$GDP_{jt} = (1 + \alpha)w_{jt}L_{jt} \quad (3.29)$$

GDP is therefore proportional to the aggregate wage bill. More importantly, sectoral value added depends on the common wage and sectoral employment alone. Since the wage does not vary across sectors within a country, differences in sectoral value added are proportional to differences in sectoral employment.

### 3.7 Household's Optimization

Given the non-homothetic CES aggregator, the household's intra-temporal problem in country  $j$  is equivalent<sup>11</sup> to:

$$\begin{aligned} \min_{\{C_{jkt}, C_{jmt}, C_{jst}\}} \sum_{k=a,m,s} P_{jkt}C_{jkt} \\ \text{s.t.} \quad \sum_{k=a,m,s} \delta_k^{1/\sigma} \left( \frac{C_{jkt}}{C_{jt}^{\varepsilon_k}} \right)^{\frac{\sigma-1}{\sigma}} = 1 \end{aligned} \quad (3.30)$$

The first-order conditions<sup>12</sup> imply sectoral consumption demand:

$$C_{jkt} = \delta_k \left( \frac{P_{jkt}}{E_{jt}} \right)^{-\sigma} C_{jt}^{\varepsilon_k(1-\sigma)}, \quad (3.31)$$

where  $E_{jt} := \sum_{k=a,m,s} P_{jkt}C_{jkt}$  denotes total consumption expenditure in country  $j$  at time  $t$ . Replacing  $E_{jt}$  with the ideal price index  $P_{jt}$  derived in Appendix A.4, the expenditure share of sector  $k$  is:

$$\begin{aligned} e_{jkt} &= \frac{P_{jkt}C_{jkt}}{P_{jt}C_{jt}} \\ &= \delta_k \left( \frac{P_{jkt}}{P_{jt}} \right)^{1-\sigma} C_{jt}^{(1-\sigma)(\varepsilon_k-1)}, \quad \forall k. \end{aligned} \quad (3.32)$$

<sup>10</sup>See Appendix A.2.

<sup>11</sup>The expenditure-minimization problem is the dual of utility maximization; the relationship between the utility function and Marshallian demand mirrors the relationship between the expenditure function and Hicksian demand.

<sup>12</sup>See Appendix A.4 for the full derivation.

Dividing  $e_{jkt}$  by  $e_{jmt}$  and using Equation (3.27) gives the expenditure share of sector  $k = a, s$  relative to manufacturing:

$$\frac{e_{jkt}}{e_{jmt}} = \frac{\delta_k}{\delta_m} \left( \frac{A_{jmt}}{A_{jkt}} \right)^{1-\sigma} C_{jt}^{(\varepsilon_k - \varepsilon_m)(1-\sigma)}, \quad k = a, s. \quad (3.33)$$

Equation (3.33) isolates the supply- and demand-side mechanisms of structural change:  $\sigma$  governs the relative-price channel, while the difference  $\varepsilon_k - \varepsilon_m$  governs the relative Engel-curve channel.

Since  $\sigma < 1$ , an increase in sector  $k$ 's relative productivity lowers its relative price and, through complementarity, its relative expenditure share. Consumers therefore respond to the resulting decline in the relative price by reallocating expenditure toward other sectors. When income elasticities differ,  $\varepsilon_k - \varepsilon_m > 0$  implies that sector  $k$ 's expenditure share rises with aggregate consumption relative to manufacturing, whereas  $\varepsilon_k - \varepsilon_m < 0$  implies the opposite. Consistent with the empirical estimates of income elasticities,  $\varepsilon_a < \varepsilon_m < \varepsilon_s$ , so the services expenditure share rises relative to manufacturing as consumption increases, while agriculture's share falls.

### 3.8 Innovation and Structural Change in a Closed Economy

Structural change can be characterized by changes in the relative importance of sectoral value added  $s_{jkt}$ , consumption expenditure  $e_{jkt}$ , and employment  $\ell_{jkt} := \frac{L_{jkt}}{L_{jt}}$  for  $k = a, m, s$ . Equations (3.28) and (3.29) imply the following equivalence.

**Proposition I** (Value added equals employment in autarky). *In the closed economy, the sectoral value-added share equals the sectoral employment share in every period:*

$$\ell_{jkt} = s_{jkt}. \quad (3.34)$$

*Proof.* Substituting Equations (3.28) and (3.29) into the definition of the value-added share gives:

$$s_{jkt} = \frac{VA_{jkt}}{GDP_{jt}} = \frac{(1+\alpha)w_{jt}L_{jkt}}{(1+\alpha)w_{jt}L_{jt}} = \frac{L_{jkt}}{L_{jt}} = \ell_{jkt}. \quad (3.35)$$

The factor  $(1+\alpha)w_{jt}$  is common across sectors and therefore cancels from the ratio. ■

The equivalence follows from the fact that, in autarky, sectoral value added is proportional to sectoral employment. Profits and wages both scale with the size of the sector, so value added and employment have the same sectoral distribution. The distinction between value added and consumption expenditure, however, remains because households can allocate current income between consumption and R&D investment.

From the final-good market clearing condition in Equation (3.15), sectoral value added can equivalently be written as:

$$VA_{jkt} = P_{jkt}C_{jkt} + \int_0^1 P_{jkt}Z_{jkt}(v)dv. \quad (3.36)$$

so that the consumption-expenditure share in sector  $k$  is:

$$e_{jkt} = \frac{VA_{jkt} - P_{jkt}Z_{jkt}}{GDP_{jt} - \sum_{k=a,m,s} P_{jkt}Z_{jkt}}, \quad (3.37)$$

where  $Z_{jkt} := \int_0^1 Z_{jkt}(v)dv$  is aggregate sector- $k$  R&D investment in country  $j$ . Dividing numerator and denominator by GDP gives the relationship between the consumption-expenditure share and the value-added share:

$$s_{jkt} = \frac{P_{jkt}Z_{jkt}}{GDP_{jt}} + \left(1 - \frac{\sum_{k=a,m,s} P_{jkt}Z_{jkt}}{GDP_{jt}}\right) e_{jkt}. \quad (3.38)$$

Thus, even in a closed economy, value-added and consumption-expenditure shares differ whenever R&D investment represents a positive share of GDP. Section 4 extends the model to international trade in intermediate goods and derives the resulting divergence between value-added and employment shares.

## 4 Opening the Economy

This section introduces trade in intermediate goods between the domestic country and the rest of the world. For simplicity, I consider two countries,  $J = \{H, F\}$ , where  $H$  denotes home and  $F$  denotes foreign. The two countries have access to the same continuum of intermediate varieties and face iceberg trade costs. Opening the economy allows final-good producers in each country to use the globally most productive version of each intermediate variety. For each variety  $v$  in sector  $k$ , the world market is therefore served by the country that holds the patent for the most advanced version. Following [Aghion and Howitt \(2009\)](#), and departing from [Eaton and Kortum \(2002\)](#), I assume that final-good producers always use the best available version of each intermediate variety, with the price determined by the patent holder. Trade costs are exogenous iceberg costs: shipping one unit of variety  $v$  of sector  $k$  from country  $i$  delivers  $\frac{1}{\tau_{ijk}}$  units in country  $j$ , with<sup>13</sup>  $\tau_{ijk} = 1 \forall k = a, m, s$ .

### 4.1 Final Goods Production

The producer of final good  $k$  uses domestic labor and a continuum of intermediate goods supplied by global patent monopolists. The productivity of the latest available version of variety  $v$  is the world frontier

$$\hat{A}_{kt}(v) := \max \{A_{Hkt}(v), A_{Fkt}(v)\}.$$

The country producing this latest version holds the patent and is the sole supplier of the variety. Thus, in country  $j$ , the final-good producer uses the world frontier productivity  $\hat{A}_{kt}(v)$  and pays the delivered price  $\hat{p}_{jkt}(v)$ , which equals the domestic monopolist's price when country  $j$  holds the leading patent and the foreign monopolist's price inclusive of trade costs otherwise. The final-good producer solves:

$$\max_{\{L_{jkt}, [x_{jkt}(v)]_{v \in [0,1]}\}} P_{jkt} L_{jkt}^{1-\alpha} \int_0^1 \hat{A}_{kt}(v)^{1-\alpha} x_{jkt}(v)^\alpha dv - \int_0^1 \hat{p}_{jkt}(v) x_{jkt}(v) dv - w_{jt} L_{jkt}, \quad (4.1)$$

<sup>13</sup>There is no trade cost within a country. Services, though traditionally considered less tradable, have become increasingly tradable with advances in digital technology; see [Lewis et al., 2021](#) and [Sposi et al., 2021](#).

where the price paid by the final-good producer in country  $j$  for the latest version of variety  $v$  is:

$$\hat{p}_{jkt}(v) = \begin{cases} p_{jkt}(v) & \text{if } \hat{A}_{kt}(v) = A_{jkt}(v) \\ \tau_{ijk} p_{ikt}(v) & \text{if } \hat{A}_{kt}(v) > A_{jkt}(v) \end{cases}$$

with  $p_{jkt}(v) = \alpha^{-1} P_{jkt}$  for all  $j \neq i \in J$ . The corresponding demands for varieties produced domestically and abroad are:

$$\begin{cases} x_{jkt}^j(v) = \left( \frac{p_{jkt}(v)}{\alpha P_{jkt}} \right)^{\frac{1}{\alpha-1}} A_{jkt}(v) L_{jkt} \\ x_{jkt}^i(v) = \left( \frac{\tau_{ijk} p_{ikt}(v)}{\alpha P_{jkt}} \right)^{\frac{1}{\alpha-1}} A_{ikt}(v) L_{jkt} \end{cases}$$

where  $x_{jkt}^j(v)$  denotes demand for a variety produced in country  $j$  and  $x_{jkt}^i(v)$  denotes demand for a variety produced in country  $i$ .

**Remark 4.1** (Why trade costs do not allocate the monopoly right). *The world market for variety  $v$  is assigned entirely to the country that holds the most advanced patent, independently of trade costs. This is the standard quality-ladder convention in which the patent holder, rather than relative prices, determines which version is available to final-good producers (Grossman and Helpman, 1991; Aghion and Howitt, 2009). It differs from the price-and-trade-cost-based competition considered by Eaton and Kortum (2002). The assumption is most natural when productivity differences between countries are sufficiently large that the technological advantage of the leading producer dominates the additional cost of serving foreign markets. A hybrid specification in which technologically inferior producers can compete on price when productivity differences are small would modify the allocation of varieties near the boundary, but not the mechanism developed below: the value-added–employment wedge depends on which country captures the monopoly rents associated with the varieties it supplies internationally.*

## 4.2 Intermediate-Good Production and World Monopoly

For each intermediate variety  $v$  in sector  $k$ , the country that holds the most advanced patent supplies the world market. Suppose country  $j$  is the technological leader for variety  $v$ . The patent holder chooses its price to maximize profits from domestic and foreign demand:

$$\begin{aligned} \max_{\{p_{jkt}(v)\}} \quad & \pi_{jkt}(v) = p_{jkt}(v) X_{kt}^j(v) - P_{jkt} X_{kt}^j(v) \\ \text{s.t.} \quad & \begin{cases} X_{kt}^j(v) = x_{jkt}^j(v) + x_{ikt}^j(v) \\ x_{jkt}^j(v) = \left( \frac{p_{jkt}(v)}{\alpha P_{jkt}} \right)^{\frac{1}{\alpha-1}} A_{jkt}(v) L_{jkt} \\ x_{ikt}^j(v) = \left( \frac{\tau_{jik} p_{jkt}(v)}{\alpha P_{ikt}} \right)^{\frac{1}{\alpha-1}} A_{jkt}(v) L_{ikt} \end{cases} \end{aligned} \tag{4.2}$$

The monopolist faces domestic and foreign demand, but the markup is unchanged because the demand elasticity is the same in both markets. Solving Equation (4.2), the monopolist sets:

$$p_{jkt}(\mathbf{v}) = \alpha^{-1} P_{jkt}, \quad (4.3)$$

Thus, the open-economy pricing rule is identical to the closed-economy rule in Equation (3.18), confirming Remark 3.1. The corresponding demands are:

$$\begin{cases} x_{jkt}^j(\mathbf{v}) = \alpha^{\frac{2}{1-\alpha}} A_{jkt}(\mathbf{v}) L_{jkt} \\ x_{ikt}^j(\mathbf{v}) = \alpha^{\frac{2}{1-\alpha}} \left( \frac{P_{ikt}}{\tau_{jik} P_{jkt}} \right)^{\frac{1}{1-\alpha}} A_{jkt}(\mathbf{v}) L_{ikt} \end{cases}$$

where the superscript  $j$  indicates that the variety is produced by the patent holder in country  $j$ .

The profit of the world monopolist producing variety  $\mathbf{v}$  in sector  $k$  in country  $j$  is therefore:

$$\begin{aligned} \pi_{jkt}(\mathbf{v}) &= (\alpha^{-1} - 1) P_{jkt} [x_{jkt}^j(\mathbf{v}) + x_{ikt}^j(\mathbf{v})] \\ &= \pi P_{jkt} A_{jkt}(\mathbf{v}) \left[ L_{jkt} + \left( \frac{P_{ikt}}{\tau_{jik} P_{jkt}} \right)^{\frac{1}{1-\alpha}} L_{ikt} \right], \end{aligned} \quad (4.4)$$

where  $\pi := (1 - \alpha) \alpha^{\frac{1+\alpha}{1-\alpha}}$ . The first term inside the brackets captures domestic demand, while the second captures foreign demand adjusted for relative prices and trade costs.

Let  $\Theta_{jkt}$  denote the set of sector- $k$  varieties for which country  $j$  holds the technological lead:

$$\int_{\Theta_{jkt}} A_{jkt}(\mathbf{v}) d\mathbf{v} = \int_0^1 A_{jkt}(\mathbf{v}) \mathbf{1}_{\{A_{jkt}(\mathbf{v}) > A_{ikt}(\mathbf{v})\}} d\mathbf{v}.$$

Aggregating profits across the varieties led by country  $j$  gives:

$$\Pi_{jkt} = \pi P_{jkt} \left[ L_{jkt} + \left( \frac{P_{ikt}}{\tau_{jik} P_{jkt}} \right)^{\frac{1}{1-\alpha}} L_{ikt} \right] \int_{\Theta_{jkt}} A_{jkt}(\mathbf{v}) d\mathbf{v}. \quad (4.5)$$

Hence, monopoly profits in sector  $k$  depend on both domestic and foreign market size. When country  $j$  leads a variety, foreign demand generates additional monopoly revenue for the domestic patent holder. By contrast, when country  $j$  does not lead any varieties in the sector, the associated monopoly rents accrue to foreign producers.

### 4.3 Innovation and Technology Leadership

Opening the economy creates an additional innovation margin because a country can gain the patent for a variety currently led by its trading partner. Each country can invest in R&D to improve its own version of an intermediate variety. A successful innovation raises the innovator's productivity by the factor  $\gamma_{jk}$ :

$$A_{jkt+1}(\mathbf{v}) = \begin{cases} \gamma_{jk} A_{jkt}(\mathbf{v}) & \text{with probability } \mu_{jkt}(\mathbf{v}) \\ A_{jkt}(\mathbf{v}) & \text{with probability } 1 - \mu_{jkt}(\mathbf{v}) \end{cases}, \quad (4.6)$$

and analogously for country  $i$ .<sup>14</sup>

Suppose country  $j$  currently leads variety  $\mathbf{v}$ , so that  $A_{jkt}(\mathbf{v}) > A_{ikt}(\mathbf{v})$ . Define the follower's relative productivity as

$$g_{jkt}(\mathbf{v}) := \frac{A_{ikt}(\mathbf{v})}{A_{jkt}(\mathbf{v})} \in (0, 1].$$

A successful innovation by the follower overtakes the incumbent whenever  $\gamma_{jk} g_{jkt}(\mathbf{v}) > 1$ . This creates two innovation regimes. In a close rivalry,  $g_{jkt}(\mathbf{v}) > 1/\gamma_{jk}$ , one successful innovation by the follower is sufficient to change the technology leader. In a far rivalry,  $g_{jkt}(\mathbf{v}) \leq 1/\gamma_{jk}$ , a single successful innovation by the follower cannot close the productivity gap. Appendix A.6 derives the optimal innovation incentives in both regimes; the main results are summarized below.

**The leader's innovation incentive.** In a close rivalry, the leader's optimal innovation probability is:

$$\mu_{jkt}^{\text{leader, close}} = \lambda_j \beta \pi \frac{P_{jkt+1}}{P_{jkt}} \left[ L_{jkt+1} + \left( \frac{P_{ikt+1}}{\tau_{jik} P_{jkt+1}} \right)^{\frac{1}{1-\alpha}} L_{ikt+1} \right] \frac{\gamma_{jk} - 1 + \mu_{ikt}(\mathbf{v})}{\gamma_{jk}}, \quad (4.7)$$

which increases with the follower's success probability. A higher probability that the follower overtakes increases the value of extending the incumbent's lead.

In a far rivalry, the follower cannot overtake in one innovation step, so the leader's innovation probability is:

$$\mu_{jkt}^{\text{leader, far}} = \lambda_j \beta \pi \frac{P_{jkt+1}}{P_{jkt}} \left[ L_{jkt+1} + \left( \frac{P_{ikt+1}}{\tau_{jik} P_{jkt+1}} \right)^{\frac{1}{1-\alpha}} L_{ikt+1} \right] (\gamma_{jk} - 1), \quad (4.8)$$

which does not depend on the follower's innovation probability. The leader therefore faces the same innovation incentive as in the closed economy, except that its potential market now includes foreign demand.

**Remark 4.2** (Rival investment and the leader's incentive). *Equations (4.7)–(4.8) show that innovation incentives are strategically interdependent only when the two countries are sufficiently close technologically. In a close rivalry, a higher follower success probability increases the leader's incentive to innovate because failing to extend the lead risks losing the associated monopoly rents. When the productivity gap is sufficiently large, this interdependence disappears because a single successful innovation by the follower cannot change technology leadership.*

**The follower's innovation incentive.** In a close rivalry, a successful innovation by the follower produces immediate technology leadership. Its optimal innovation probability is:

$$\mu_{ikt}^{\text{follower, close}} = \lambda_i \beta \pi \frac{P_{ikt+1}}{P_{ikt}} \left[ L_{ikt+1} + \left( \frac{P_{jkt+1}}{\tau_{ijk} P_{ikt+1}} \right)^{\frac{1}{1-\alpha}} L_{jkt+1} \right] (1 - \mu_{jkt}(\mathbf{v})), \quad (4.9)$$

which decreases with the incumbent's innovation probability because the follower benefits from successful innovation only when the incumbent fails to extend its lead. Thus, close technological competition

<sup>14</sup>This assumes a common step size  $\gamma_{jk} = \gamma_{ik}$  within a sector: the size of an innovation step is a property of the sector's technology, not of which country achieves it.

generates strategic interaction in both directions.

In a far rivalry, a successful follower innovation cannot change technology leadership within the current period. Under the one-period innovation payoff used in the model, the follower therefore has no immediate return from innovation and its optimal innovation probability is zero.<sup>15</sup>

Relative to the closed economy, a technology leader now has access to a larger effective market. Comparing Equations (4.7) and (4.8) with Equation (3.22), the leader's innovation incentive incorporates foreign employment through the additional demand for its patented varieties. Trade therefore increases the return to maintaining technological leadership whenever the foreign market is accessible. The follower's innovation incentive is a new margin created by international competition: a domestic entrepreneur can invest in a variety currently led abroad in order to become the global patent holder.

#### 4.4 Aggregate Productivity and Market Clearing

The first-order conditions of Equation (4.1) imply the wage equation:

$$w_{jt} = (1 - \alpha) \alpha^{\frac{2\alpha}{1-\alpha}} P_{jkt} A_{jkt}, \quad (4.10)$$

where aggregate productivity available to final-good producers in country  $j$  is:

$$A_{jkt} = \int_{\Theta_{jkt}} A_{jkt}(v) dv + \left( \frac{P_{jkt}}{\tau_{ijk} P_{ikt}} \right)^{\frac{\alpha}{1-\alpha}} \int_{\Theta_{ikt}} A_{ikt}(v) dv. \quad (4.11)$$

The first term captures productivity from domestically led varieties, while the second captures the productivity embodied in imported varieties after accounting for trade costs and relative prices. Hence, foreign technological progress can raise productivity in country  $j$  by improving the quality of the intermediate varieties available to its final-good producers.

Final-good market clearing in country  $j$  and sector  $k$  requires:

$$Y_{jkt} = C_{jkt} + Z_{jkt} + \int_{\Theta_{jkt}} \left[ x_{jkt}^j(v) + x_{ikt}^j(v) \right] dv. \quad (4.12)$$

The three terms on the right-hand side represent household consumption, R&D investment, and production of intermediate varieties by domestic patent holders for domestic and foreign final-good producers, respectively. Thus, the final good produced in country  $j$  finances both domestic uses and the production of intermediate varieties that generate monopoly revenue from foreign demand.

#### 4.5 Equilibrium and Balanced Trade

**Definition 2** (Open-economy equilibrium). *Given the technology state  $\{A_{jkt}(v), A_{ikt}(v)\}_{v \in [0,1]}$  and the resulting leadership sets  $\{\Theta_{jkt}, \Theta_{ikt}\}$  inherited from Step 0 of the timing protocol, an open-economy equilibrium for period  $t$  consists of wages  $(w_{Ht}, w_{Ft})$ , sectoral prices  $\{P_{Hkt}, P_{Fkt}\}_{k=a,m,s}$ , consumption and R&D allocations, and intermediate-input demands in both countries such that final-good producers*

<sup>15</sup>A distant follower can still have a continuation value from reducing the productivity gap. Capturing that value requires the multi-period, state-contingent framework of Aghion et al. (2001). The zero innovation result here therefore applies only to a rivalry that begins sufficiently far apart that one innovation step cannot change leadership. It does not characterize followers in general and does not affect Proposition II, which depends only on the technology-leadership partition  $\Theta_{jkt}$ .

solve Equation (4.1), intermediate monopolists solve Equation (4.2), households solve Equation (3.9), and labor and final-good markets clear in both countries according to Equations (3.14) and (4.12).

**Lemma 4.1** (Existence). *For given trade costs  $\{\tau_{ijk}\}$ , population sizes  $\{L_{Ht}, L_{Ft}\}$ , and technology state  $\{A_{jkt}(v)\}$ , a within-period open-economy equilibrium exists.*

*Proof sketch.* See Appendix A.5. Conditional on the technology state and leadership sets, the within-period equilibrium is a standard competitive equilibrium with monopolistically supplied intermediate varieties. The excess-demand system is continuous and homogeneous of degree zero in prices. A fixed-point argument on a normalized price simplex therefore delivers a within-period equilibrium. This result is conditional on the inherited technology state and does not establish existence of a dynamic sequential equilibrium in which the leadership sets evolve endogenously through the innovation process. ■

The absence of international asset markets imposes an additional restriction on the equilibrium that is useful for the analysis of the value-added–employment wedge.

**Lemma 4.2** (Balanced trade). *In every period  $t$ , a country's aggregate net exports of intermediate varieties across all sectors are identically zero:*

$$\sum_{k=a,m,s} NX_{jkt} = 0, \quad (4.13)$$

where  $NX_{jkt}$  is defined in Equation (4.20).

*Proof.* The household's budget constraint in Equation (3.3) binds at the optimum, since utility is strictly increasing in  $C_{jt}$ :

$$\sum_{k=a,m,s} P_{jkt}(C_{jkt} + Z_{jkt}) = w_{jt}L_{jt} + \sum_{k=a,m,s} \Pi_{jkt}. \quad (4.14)$$

GDP measured by the income approach is the sum of labor income and domestic monopoly profits, regardless of whether the underlying intermediate-good sales occur at home or abroad:

$$GDP_{jt} = w_{jt}L_{jt} + \sum_{k=a,m,s} \Pi_{jkt}. \quad (4.15)$$

Summing the sectoral value-added identity in Equation (4.19) over sectors gives:

$$GDP_{jt} = \sum_{k=a,m,s} P_{jkt}(C_{jkt} + Z_{jkt}) + \sum_{k=a,m,s} NX_{jkt}. \quad (4.16)$$

Substituting Equation (4.14) into Equation (4.16) and using Equation (4.15) yields:

$$GDP_{jt} = GDP_{jt} + \sum_{k=a,m,s} NX_{jkt}, \quad (4.17)$$

which implies Equation (4.13). ■

Balanced trade does not imply that trade is irrelevant for the distribution of value added across sectors. A country can export intermediate varieties in sectors in which it holds the global technology lead and import varieties in sectors led by foreign producers. The resulting sectoral trade imbalances are

offset across sectors, so aggregate net exports are zero, while the location of monopoly rents continues to differ across sectors. The value-added–employment wedge derived below therefore arises from the location of these rents rather than from aggregate borrowing or lending.

The balanced-trade restriction follows from the absence of an international asset market: countries cannot finance a current-account deficit or accumulate foreign claims within the period. The model therefore isolates the cross-sector mechanism through which international trade changes the location of value added. In particular, access to foreign technology affects domestic productivity and wages through Equation (4.11), while the distribution of monopoly rents across countries affects sectoral value added. Introducing international asset markets would allow persistent trade deficits or surpluses and would address a different question, namely how capital flows interact with structural transformation over time, as in Kehoe et al. (2018). The period-by-period balanced-trade assumption is also used by Uy et al. (2013) in a related setting.

#### 4.6 Monopoly Rights, Trade, and Structural Change

This subsection establishes the relationships between sectoral value-added and consumption-expenditure shares and between sectoral value-added and employment shares. It shows how international trade and the resulting distribution of monopoly rights modify the closed-economy results of Section 3.

**Consumption Expenditure and Value-Added Shares.** The value added of sector  $k$  in country  $j$  is:

$$\begin{aligned}
 VA_{jkt} = & P_{jkt} Y_{jkt} - \underbrace{\int_{\Theta_{jkt}} p_{jkt}(v) x_{jkt}^j(v) dv - \int_{\Theta_{ikt}} \tau_{ijk} p_{ikt}(v) x_{jkt}^i(v) dv}_{\text{Final sector value added}} \\
 & + \underbrace{\int_{\Theta_{jkt}} (p_{jkt}(v) - P_{jkt}) [x_{jkt}^j(v) + x_{ikt}^j(v)] dv}_{\text{Intermediate varieties value added}}. \tag{4.18}
 \end{aligned}$$

Using final-goods market clearing, Equation (4.12), Equation (4.18) becomes:

$$VA_{jkt} = P_{jkt} (C_{jkt} + Z_{jkt}) + NX_{jkt}, \tag{4.19}$$

where  $NX_{jkt}$  is net exports of all varieties of sector  $k$  for country  $j$  at period  $t$ :

$$NX_{jkt} = \int_{\Theta_{jkt}} p_{jkt}(v) x_{ikt}^j(v) dv - \int_{\Theta_{ikt}} \tau_{ijk} p_{ikt}(v) x_{jkt}^i(v) dv. \tag{4.20}$$

From Equation (4.19), the consumption-expenditure share in sector  $k$  is:

$$e_{jkt} = \frac{VA_{jkt} - NX_{jkt} - P_{jkt} Z_{jkt}}{GDP_{jt} - \sum_{k=a,m,s} P_{jkt} Z_{jkt} - \sum_{k=a,m,s} NX_{jkt}}. \tag{4.21}$$

Dividing numerator and denominator by  $GDP_{jt}$  and rearranging gives:

$$s_{jkt} = \left( 1 - \frac{\sum_{k=a,m,s} P_{jkt} Z_{jkt}}{GDP_{jt}} - \frac{\sum_{k=a,m,s} NX_{jkt}}{GDP_{jt}} \right) e_{jkt} + \frac{P_{jkt} Z_{jkt} + NX_{jkt}}{GDP_{jt}}. \tag{4.22}$$

Lemma 4.2 then removes the aggregate net-export term from the expression.

**Corollary 4.1** (Simplified value-added-expenditure identity). *Under balanced trade,*

$$s_{jkt} = \left(1 - \frac{\sum_{n=a,m,s} P_{jnt} Z_{jnt}}{GDP_{jt}}\right) e_{jkt} + \frac{P_{jkt} Z_{jkt} + NX_{jkt}}{GDP_{jt}}. \quad (4.23)$$

*Proof.* Substitute  $\sum_{k=a,m,s} NX_{jkt} = 0$ , from Lemma 4.2, into Equation (4.22). ■

Equation (4.23) shows that, under balanced trade, the difference between a sector's value-added and consumption-expenditure shares reflects its own R&D expenditure and net exports, relative to an economy-wide R&D adjustment. The aggregate trade balance does not enter because it is zero by construction. Thus, holding the economy-wide R&D expenditure share fixed, a sector with higher net exports or higher R&D expenditure has a higher value-added share relative to its consumption-expenditure share, while the opposite holds for a sector with lower net exports or lower R&D expenditure.

**Employment and Value-Added Shares.** Value added in sector  $k$  also equals the sum of wages and profits paid in that sector:

$$VA_{jkt} = w_{jt} L_{jkt} + \Pi_{jkt}. \quad (4.24)$$

Using Equation (4.5) for  $\Pi_{jkt}$ , the value-added share is:

$$s_{jkt} = \frac{w_{jt} L_{jkt} + \pi P_{jkt} \left[ L_{jkt} + \left( \frac{P_{ikt}}{\tau_{jik} P_{jkt}} \right)^{\frac{1}{1-\alpha}} L_{ikt} \right] \int_{\Theta_{jkt}} A_{jkt}(\nu) d\nu}{w_{jt} L_{jt} + \sum_{n=a,m,s} \pi P_{jnt} \left[ L_{jnt} + \left( \frac{P_{int}}{\tau_{jin} P_{jnt}} \right)^{\frac{1}{1-\alpha}} L_{int} \right] \int_{\Theta_{jnt}} A_{jnt}(\nu) d\nu}. \quad (4.25)$$

Dividing numerator and denominator by  $w_{jt} L_{jt}$  and expressing the result in terms of the employment share  $\ell_{jkt} := \frac{L_{jkt}}{L_{jt}}$  gives:

$$\frac{s_{jkt}}{\ell_{jkt}} = \frac{1 + \alpha \left[ 1 + \left( \frac{P_{ikt}}{\tau_{jik} P_{jkt}} \right)^{\frac{1}{1-\alpha}} \frac{L_{ikt}}{L_{jkt}} \right] \frac{1}{A_{jkt}} \int_{\Theta_{jkt}} A_{jkt}(\nu) d\nu}{1 + \alpha \sum_{n=a,m,s} \ell_{jnt} \left[ 1 + \left( \frac{P_{int}}{\tau_{jin} P_{jnt}} \right)^{\frac{1}{1-\alpha}} \frac{L_{int}}{L_{jnt}} \right] \frac{1}{A_{jnt}} \int_{\Theta_{jnt}} A_{jnt}(\nu) d\nu}. \quad (4.26)$$

Equation (4.26) provides an exact expression for the value-added–employment share ratio. Its economic content becomes clearer when it is expressed in terms of three sector-level indices that separately capture technological leadership, foreign-demand exposure, and their interaction.

**Definition 3** (Leadership, exposure, and profit-intensity indices). *For country  $j$ , sector  $k$ , and period  $t$ , define:*

$$\theta_{jkt} := \frac{1}{A_{jkt}} \int_{\Theta_{jkt}} A_{jkt}(\nu) d\nu \in [0, 1], \quad (4.27)$$

$$\phi_{jkt} := \left( \frac{P_{ikt}}{\tau_{jik} P_{jkt}} \right)^{\frac{1}{1-\alpha}} \frac{L_{ikt}}{L_{jkt}} \geq 0, \quad (4.28)$$

$$\psi_{jkt} := \theta_{jkt} (1 + \phi_{jkt}). \quad (4.29)$$

$\theta_{jkt}$  is the share of sector  $k$ 's aggregate productivity embodied in varieties for which country  $j$  holds the global technology lead.  $\phi_{jkt}$  measures foreign relative to domestic demand for these led varieties, adjusted for relative prices and trade costs. The product  $\psi_{jkt}$  summarizes the resulting profit intensity of sector  $k$ .

That  $\theta_{jkt} \in [0, 1]$  follows from Equation (4.11):  $A_{jkt}$  is the sum of  $\int_{\Theta_{jkt}} A_{jkt}(v)dv$  and a non-negative term, so the first cannot exceed  $A_{jkt}$ .

**Proposition II** (Value-added-employment wedge). *Equation (4.26) is equivalent to*

$$\frac{s_{jkt}}{\ell_{jkt}} = \frac{1 + \alpha \psi_{jkt}}{1 + \alpha \bar{\psi}_{jt}}, \quad \bar{\psi}_{jt} := \sum_{n=a,m,s} \ell_{jnt} \psi_{jnt}, \quad (4.30)$$

where  $\bar{\psi}_{jt}$  is the employment-share-weighted average profit-intensity index across sectors of country  $j$ . Consequently,

$$s_{jkt} \geq \ell_{jkt} \iff \psi_{jkt} \geq \bar{\psi}_{jt}. \quad (4.31)$$

*Proof.* Substituting Equations (4.27)-(4.29) into the numerator of Equation (4.26) gives  $1 + \alpha \theta_{jkt}(1 + \phi_{jkt}) = 1 + \alpha \psi_{jkt}$ , and into the denominator gives  $1 + \alpha \sum_n \ell_{jnt} \theta_{jnt}(1 + \phi_{jnt}) = 1 + \alpha \bar{\psi}_{jt}$ . Since  $1 + \alpha \bar{\psi}_{jt} > 0$ , the sign equivalence in Equation (4.31) follows directly from Equation (4.30). ■

Proposition II states the paper's central result in one line: a sector's value-added share exceeds its employment share exactly when its profit intensity exceeds the country's economy-wide average. Profit intensity depends on both the country's leadership over sectoral intermediate varieties and the demand for those varieties abroad. Technological leadership determines which varieties generate monopoly rents for country  $j$ , while foreign demand determines the additional scale of those rents. A sector with little or no technological leadership therefore has low profit intensity regardless of its foreign-demand exposure, whereas foreign demand amplifies the profit intensity of sectors in which country  $j$  holds monopoly rights.

**Corollary 4.2** (Autarky nesting). *As the foreign country's population in every sector vanishes,  $L_{ikt} \rightarrow 0 \forall k$ , the open-economy expressions converge to the closed-economy case. In particular,  $\Theta_{jkt} \rightarrow [0, 1]$ , so  $\theta_{jkt} \rightarrow 1$ , while  $\phi_{jkt} \rightarrow 0$  directly from Equation (4.28). Hence  $\psi_{jkt} \rightarrow 1$  for every sector  $k$ ,  $\bar{\psi}_{jt} \rightarrow 1$ , and Equation (4.30) collapses to  $s_{jkt}/\ell_{jkt} = 1$  for every  $k$ , recovering Proposition I.*

Corollary 4.2 shows that the closed-economy equivalence of Section 3 is the limiting case of Proposition II. With a single country, every sector's varieties are domestically led and there is no foreign demand, so every sector has the same profit-intensity index,  $\psi_{jkt} = 1$ . The value-added-employment wedge therefore vanishes identically.

The structure of the foreign economy also shapes the wedge through  $\phi_{jkt}$ . Holding relative prices and trade costs fixed, if sector  $k$  is relatively larger abroad than at home compared with sector  $n$ , in the sense that  $L_{ikt}/L_{jkt} > L_{int}/L_{jnt}$ , then  $\phi_{jkt} > \phi_{jnt}$ . Holding  $\theta$  fixed, sector  $k$  therefore has greater profit intensity and a larger value-added share relative to employment than sector  $n$ . This transmission channel operates through demand for intermediate goods: a larger foreign market for a sector increases demand for the varieties that country  $j$ 's firms lead in that sector, raising their sales, profits, and value added.

Similarly, a higher relative foreign price net of trade costs,  $P_{ikt}/(\tau_{ijk}P_{jkt}) > P_{int}/(\tau_{ijn}P_{jnt})$ , raises  $\phi_{jkt}$  relative to  $\phi_{jnt}$  and therefore increases sector  $k$ 's profit intensity relative to sector  $n$ 's, holding  $\theta$  fixed.

**Corollary 4.3** (Technological Leadership and the Wedge). *If country  $j$  leads no variety of sector  $k$ , so that  $\theta_{jkt} = 0$ , then  $\psi_{jkt} = 0$  regardless of  $\phi_{jkt}$ , and by Equation (4.31),  $s_{jkt} < \ell_{jkt}$  whenever  $\bar{\psi}_{jt} > 0$ , i.e., whenever the country leads at least one variety in some sector.*

*Proof.* Immediate from Equations (4.29) and (4.31) with  $\theta_{jkt} = 0$ . ■

Corollary 4.3 provides a direct implication for the sectoral pattern documented in Figure 1. Agriculture's value-added share falls short of its employment share in 98.1% of country-sector-year observations, substantially more consistently than manufacturing or services, and this pattern holds across 42 countries. The corollary shows that a sector need not be a net importer of intermediate goods to exhibit a negative wedge. If a country holds no monopoly rights over the varieties used in a sector, then  $\theta_{jkt} = 0$  and its profit intensity is zero, so its employment share exceeds its value-added share whenever the country earns monopoly rents in at least one other sector.

This mechanism is consistent with the concentration of research and development across sectors. Such uniformity points to a structural explanation rather than a country-specific one. Manufacturing is consistently classified among the most research-intensive sectors in nearly every economy, while agriculture sits persistently among the least research-intensive, a regularity documented across OECD countries by the R&D-intensity taxonomy of Galindo-Rueda and Verger (2016). Because holding global technological leadership in a sector requires sustained research investment to stay ahead of foreign rivals, this same concentration implies that  $\theta_{jkt}$  is structurally low for agriculture and structurally higher for manufacturing in nearly every country, not merely in some: agricultural intermediate varieties are less concentrated in globally-monopolized, patent-protected technology than manufacturing varieties to begin with (Comin et al., 2025). This is why the pattern in Figure 1 is not merely common but nearly universal: it reflects where research and development itself is concentrated, a fact about the technology of innovation rather than about any single country's trade position.

The mechanism does not, however, require technological innovation to be the source of the rent-generating advantage. Control over a scarce natural resource can generate an analogous position when the resource is a valuable intermediate input demanded abroad. The key mechanism is the international capture of rents associated with a valuable intermediate input. In the model, technological leadership gives domestic firms exclusive control over advanced varieties and allows them to earn rents from domestic and foreign demand. Similarly, control over a scarce natural resource can allow a country to capture rents from foreign demand without a proportional increase in domestic employment. What matters is that the income generated by the valuable intermediate input accrues to the country that controls its supply.

This perspective also extends the interpretation of the wedge beyond income differences across countries. A country need not be close to the technological frontier to exhibit a positive value-added–employment share gap if it controls a scarce and internationally demanded input. Resource-rich developing countries, for example, can capture substantial rents through foreign demand even when technological leadership in other sectors is concentrated abroad. More generally, the gap can reflect differences in the distribution of rent-generating advantages across sectors, so countries at similar levels of development can exhibit different gaps when their sources of rents differ, while countries at different income

levels can exhibit similar gaps when they occupy comparable positions in international markets for valuable intermediate inputs.

**Corollary 4.4** (Complete Technological Followership). *If a country leads no variety in any sector, so that  $\theta_{jkt} = 0$  for every  $k$ , then  $\psi_{jkt} = 0$  for every  $k$ ,  $\bar{\psi}_{jt} = 0$ , and by Equation (4.30),  $s_{jkt}/\ell_{jkt} = 1$  in every sector: the value-added-employment wedge vanishes exactly as under autarky.*

*Proof.* Immediate from Equations (4.29) and (4.30) with  $\theta_{jkt} = 0$  for every  $k$ . ■

Corollary 4.4 clarifies the scope of the mechanism: the wedge arises from differences in domestic monopoly-rent capture across sectors, not from trade openness alone. A country that imports every variety in every sector still gains access to foreign technology, which affects domestic productivity and wages through Equation (4.11). But because the country holds no monopoly rights in any sector,  $\psi_{jkt} = 0$  for every  $k$ , so there is no cross-sector variation in profit intensity and the value-added-employment wedge vanishes. The wedge therefore requires uneven technological leadership across a country's sectors. It vanishes at both extremes: under autarky, where every sector is domestically led (Corollary 4.2), and under complete technological followership, where no sector is domestically led.

## 5 Conclusion

This paper starts from a simple empirical puzzle: sectoral employment and value-added shares, although both measures of domestic production, need not coincide. The motivating evidence shows that this gap differs systematically across sectors and varies with international trade in intermediate goods. The same trade margins are also related to the profit share of value added, pointing to a connection between the gap and the location of production rents. The paper develops a Schumpeterian model of innovation and trade that provides a mechanism linking these patterns.

The model establishes a sharp distinction between the closed- and open-economy cases. In a closed economy, value-added and employment shares coincide because sectoral profits scale with sectoral employment, so the distribution of employment across sectors determines the distribution of value added. Opening the economy breaks this equivalence when countries differ in technological leadership across intermediate varieties. The key mechanism is the international location of monopoly rents: technological leaders capture rents from foreign as well as domestic demand for their intermediate varieties, whereas countries that rely on foreign varieties do not. Trade can therefore separate the location of employment from the location of value added without any change in the monopoly mark-up. The wedge reflects where rents are earned, not a change in the degree of market power.

The theoretical results characterize this mechanism precisely. A sector's value-added share exceeds its employment share when its profit intensity exceeds the economy-wide average. Profit intensity combines domestic technological leadership with the scale of foreign demand for the varieties over which domestic firms hold monopoly rights. The framework therefore identifies a specific source of the value-added-employment wedge: international trade changes the market over which monopoly rents can be earned and, consequently, changes where those rents enter national value added. The wedge vanishes both under autarky, when all varieties are domestically led, and under complete technological followership, when no domestic sector captures monopoly rents. Absent international borrowing, aggregate net

exports of intermediate varieties are zero in equilibrium, so the mechanism does not rely on aggregate trade imbalances or international asset positions.

These results distinguish two conceptually different ways in which international trade can affect structural transformation. Trade in final goods can separate domestic production from domestic consumption, as emphasized in the existing structural-transformation literature. Trade in intermediate varieties creates a different margin: it can separate value added from employment within domestic production by changing the international location of monopoly rents. This distinction provides a unified link between intermediate-goods trade, technological leadership, sectoral profits, and the value-added–employment share gap. It also shows that the wedge need not be interpreted solely through technological innovation. The same rent-location mechanism can apply when control over a scarce and internationally demanded intermediate input allows a country to capture rents from foreign demand.

The empirical patterns documented at the outset are consistent with this mechanism. Sectors with greater intermediate-good export exposure tend to have larger value-added–employment gaps and higher profit shares, whereas intermediate-good imports are associated with the opposite pattern. The model provides a theoretical interpretation of these relationships by identifying the location of monopoly rents as the link between international trade and the distribution of value added across sectors. In this sense, the paper shifts attention from structural change measured solely through employment, value added, or consumption shares toward the international allocation of the rents embedded in intermediate production.

The framework also raises a broader question about structural transformation in an integrated world economy: how does a country’s position in the international distribution of rents shape the allocation of employment and value added across sectors? The results suggest that technological leadership and control over valuable intermediate inputs may matter not only for productivity and income, but also for how structural change is measured and interpreted. Establishing whether changes in international rent capture systematically shape the evolution of sectoral employment and value added across countries is an important direction for future research.

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## A Mathematical Appendix

### A.1 Goods production sectors

A drastic innovation gives its holder the entire market for variety  $v$ , as defined in Section 3. The incumbent producer of variety  $v$  in sector  $k$  of country  $j$  therefore acts as an unconstrained monopolist. It chooses the quantity  $x_{jkt}(v)$  supplied to the sector- $k$  final-good producer, taking the inverse demand implied by the final-good producer's cost-minimization problem as given. The problem of intermediate firm  $j_k$  with drastic innovation is:

$$\begin{aligned} \max_{\{x_{jkt}(v)\}} \quad & \pi_{jkt}(v) = p_{jkt}(v)x_{jkt}(v) - P_{jkt}x_{jkt}(v) \\ \text{s.t.} \quad & p_{jkt}(v) = \alpha P_{jkt}x_{jkt}(v)^{\alpha-1}A_{jkt}(v)^{1-\alpha}L_{jkt}^{1-\alpha} \end{aligned} \quad (\text{A.1})$$

Substituting the inverse demand into the objective leaves an unconstrained choice of  $x_{jkt}(v)$ . The first-order condition is:

$$\alpha^2 P_{jkt}x_{jkt}(v)^{\alpha-1}A_{jkt}(v)^{1-\alpha}L_{jkt}^{1-\alpha} - P_{jkt} = 0 \iff x_{jkt}(v) = \alpha^{\frac{2}{1-\alpha}}A_{jkt}(v)L_{jkt}$$

Because all varieties, sectors, and countries share the same demand curvature  $\alpha$ , the optimal quantity is proportional to the variety-specific productivity level  $A_{jkt}(v)$ . Substituting the optimal quantity into the inverse demand gives:

$$p_{jkt}(v) = \alpha^{-1}P_{jkt} \quad (\text{A.2})$$

Thus, every intermediate variety is sold at the same markup  $\alpha^{-1}$  over the final-good price. The markup is common across varieties, sectors, and countries and therefore does not generate variation in the relative value-added or employment shares studied in the main text.

### A.2 Aggregate behavior

The value added of sector  $k$  in country  $j$  in the closed economy is:

$$\begin{aligned} VA_{jkt} &= \underbrace{P_{jkt}Y_{jkt} - \int_0^1 p_{jkt}(v)x_{jkt}(v)dv}_{\text{Final sector value added}} + \underbrace{\int_0^1 (p_{jkt}(v)x_{jkt}(v) - P_{jkt}x_{jkt}(v))dv}_{\text{Intermediate varieties value added}} \\ &= P_{jkt}Y_{jkt} - \int_0^1 P_{jkt}x_{jkt}(v)dv \\ &= \alpha^{\frac{2\alpha}{1-\alpha}}P_{jkt}A_{jkt}L_{jkt} - \alpha^{\frac{2}{1-\alpha}}P_{jkt}A_{jkt}L_{jkt} \\ &= (1 - \alpha^2)\alpha^{\frac{2\alpha}{1-\alpha}}P_{jkt}A_{jkt}L_{jkt} = (1 + \alpha)w_{jt}L_{jkt} \end{aligned}$$

Hence, sectoral value added is proportional to  $P_{jkt}A_{jkt}L_{jkt}$ , with the proportionality factor  $\zeta := (1 - \alpha^2)\alpha^{2\alpha/(1-\alpha)}$ . Using  $w_{jt} = \omega P_{jkt}A_{jkt}$  and  $\zeta/\omega = 1 + \alpha$ , the same expression can be written as  $(1 + \alpha)w_{jt}L_{jkt}$ . Because  $\zeta$  is common across sectors and countries, it does not affect ratios of value added. This property is used below to verify the GDP identities.

❖ *Calculation of GDP by income perspective*

$$GDP_{jt} = w_{jt}L_{jt} + \sum_{k=a,m,s} \Pi_{jkt} \quad (\text{A.3})$$

where

$$\Pi_{jkt} = \int_0^1 \pi_{jkt}(v) dv$$

is total profit in sector  $k$ 's intermediate varieties. This identity follows from the income-accounting decomposition of GDP into labor income and profits earned by domestic intermediate-good producers. It applies in both the closed and open economies; the difference is that  $\Pi_{jkt}$  includes only domestic sales in the closed economy and both domestic and foreign sales in the open economy.

Because the wage is common across sectors, Equation (3.26) implies that  $P_{jnt}A_{jnt}$  is the same across sectors. Thus, when summing over sectors, this common term can be factored out and expressed using any sectoral index  $k$ . Substituting the closed-economy expressions for  $w_{jt}$  and  $\pi_{jkt}(v)$ :

$$\begin{aligned} GDP_{jt} &= (1 - \alpha)\alpha^{\frac{2\alpha}{1-\alpha}} P_{jkt}A_{jkt}L_{jkt} + (1 - \alpha)\alpha^{\frac{1+\alpha}{1-\alpha}} \sum_{n=a,m,s} P_{jnt}A_{jnt}L_{jnt} \\ &= (1 - \alpha)\alpha^{\frac{2\alpha}{1-\alpha}} P_{jkt}A_{jkt} + (1 - \alpha)\alpha^{\frac{1+\alpha}{1-\alpha}} P_{jkt}A_{jkt} \sum_{n=a,m,s} L_{jnt} \\ &= (1 - \alpha)\alpha^{\frac{2\alpha}{1-\alpha}} \left[ 1 + \alpha^{\frac{1+\alpha-2\alpha}{1-\alpha}} \right] P_{jkt}A_{jkt}L_{jt} \\ &= \zeta P_{jkt}A_{jkt}L_{jt} = (1 + \alpha)w_{jt}L_{jt} \end{aligned} \quad (\text{A.4})$$

where  $\zeta := (1 - \alpha^2)\alpha^{\frac{2\alpha}{1-\alpha}}$ . The final equality follows from  $w_{jt} = \omega P_{jkt}A_{jkt}$  and  $\zeta/\omega = 1 + \alpha$ .

❖ *Calculation of GDP by value-added perspective*

$$\begin{aligned} GDP_{jt} &= \sum_{n=a,m,s} VA_{jnt} \\ &= \sum_{n=a,m,s} \zeta P_{jnt}A_{jnt}L_{jnt} \\ &= \zeta P_{jkt}A_{jkt} \sum_{n=a,m,s} L_{jnt} \\ &= \zeta P_{jkt}A_{jkt}L_{jt} = (1 + \alpha)w_{jt}L_{jt} \end{aligned} \quad (\text{A.5})$$

The two calculations therefore give the same GDP expression. The equality follows from the common wage across sectors and provides the accounting consistency underlying the sectoral value-added shares used in the main text.

### A.3 Dynamics of productivity

The productivity growth rate  $g_{A_{kt}}$  of sector  $k$  follows from integrating over the continuum of varieties and applying the law of large numbers. The stochastic innovation outcome for each variety can therefore

be represented by the corresponding success probability  $\mu_{jkt}$  at the sector level:

$$\begin{aligned}
g_{A_{jkt}} &= \frac{A_{jkt+1} - A_{jkt}}{A_{jkt}} \\
&= \frac{1}{A_{jkt}} \int_0^1 \mu_{jkt} (\gamma_{jk} A_{jkt}(v) - A_{jkt}(v)) dv \\
&= \frac{\mu_{jkt}(\gamma_{jk} - 1)}{A_{jkt}} \int_0^1 A_{jkt}(v) dv \\
&= \mu_{jkt}(\gamma_{jk} - 1)
\end{aligned} \tag{A.6}$$

The common success probability and innovation step size factor out of the integral, while the remaining integral equals aggregate sectoral productivity. Hence, sectoral productivity growth is deterministic at the aggregate level and equals the innovation success probability multiplied by the productivity step size. In particular, productivity growth is strictly positive whenever  $\mu_{jkt} > 0$ . This sector-level growth rate is the productivity-growth object used in the market-size and payoff expressions derived in Appendix A.6.

#### A.4 Households' optimization

This appendix derives the household's sectoral demand system in Equation (A.13) and the expenditure shares in Equations (A.20)–(A.21). The derivation first eliminates the Lagrange multiplier from the first-order conditions and then expresses sectoral expenditure shares in terms of relative prices and aggregate consumption. The Lagrangian of the household's problem in country  $j$  is:

$$\mathcal{L}(C_{jat}, C_{jmt}, C_{jst}; \eta_j) = \sum_{k=a,m,s} P_{jkt} C_{jkt} + \eta_j \left[ 1 - \sum_{k=a,m,s} \delta_k^{1/\sigma} \left( \frac{C_{jkt}}{C_{jt}^{\epsilon_k}} \right)^{\frac{\sigma-1}{\sigma}} \right] \tag{A.7}$$

where  $\eta_j$  is the multiplier on the aggregate utility constraint. The first-order conditions are:

$$\frac{\partial \mathcal{L}}{\partial C_{jkt}} = P_{jkt} - \eta_j \delta_k^{1/\sigma} \left( \frac{\sigma-1}{\sigma} \right) \frac{C_{jt}^{\epsilon_k}}{C_{jt}^{2\epsilon_k}} \left( \frac{C_{jkt}}{C_{jt}^{\epsilon_k}} \right)^{-1/\sigma} = 0 \quad \forall k = a, m, s \tag{A.8}$$

Solving Equation (A.8) for  $P_{jkt}$  gives:

$$P_{jkt} = \eta_j \left( \frac{\sigma-1}{\sigma} \right) \frac{\delta_k^{1/\sigma}}{C_{jt}^{\epsilon_k}} \left( \frac{C_{jkt}}{C_{jt}^{\epsilon_k}} \right)^{-\frac{1}{\sigma}} \tag{A.9}$$

Multiplying by  $C_{jkt}$  gives sector- $k$  expenditure:

$$P_{jkt} C_{jkt} = \eta_j \left( \frac{\sigma-1}{\sigma} \right) \delta_k^{1/\sigma} \left( \frac{C_{jkt}}{C_{jt}^{\epsilon_k}} \right)^{1-\frac{1}{\sigma}} \quad \forall k \tag{A.10}$$

Summing across sectors and using the utility constraint yields total expenditure  $E_{jt} := \sum_{k=a,m,s} P_{jkt} C_{jkt}$ :

$$E_{jt} = \eta_j \left( \frac{\sigma-1}{\sigma} \right) \tag{A.11}$$

The multiplier can therefore be eliminated in favor of total expenditure. Rewriting Equation (A.9) gives:

$$\frac{P_{jkt}}{E_{jt}} = \delta_k^{1/\sigma} C_{jkt}^{-1/\sigma} C_{jt}^{\varepsilon_k(\frac{1}{\sigma}-1)} \quad (\text{A.12})$$

so that:

$$C_{jkt} = \delta_k \left( \frac{P_{jkt}}{E_{jt}} \right)^{-\sigma} C_{jt}^{\varepsilon_k(1-\sigma)} \quad \forall k = a, m, s \quad (\text{A.13})$$

This is the sectoral demand system used in the main text. To express it in terms of the aggregate price index and expenditure shares, raise Equation (A.9) to the power  $1 - \sigma$ :

$$P_{jkt}^{1-\sigma} = \delta_k^{\frac{1-\sigma}{\sigma}} E_{jt}^{1-\sigma} C_{jt}^{(\sigma-1)\varepsilon_k} \left( \frac{C_{jkt}}{C_{jt}^{\varepsilon_k}} \right)^{1-\frac{1}{\sigma}} \quad \forall k \quad (\text{A.14})$$

so that

$$\delta_k P_{jkt}^{1-\sigma} C_{jt}^{(1-\sigma)\varepsilon_k} = E_{jt}^{1-\sigma} \delta_k^{\frac{1}{\sigma}} \left( \frac{C_{jkt}}{C_{jt}^{\varepsilon_k}} \right)^{1-\frac{1}{\sigma}} \quad \forall k = a, m, s \quad (\text{A.15})$$

Summing Equation (A.15) across sectors and applying the utility constraint gives:

$$\sum_k \delta_k P_{jkt}^{1-\sigma} C_{jt}^{(1-\sigma)\varepsilon_k} = E_{jt}^{1-\sigma} \quad (\text{A.16})$$

$$\implies C_{jt}^{1-\sigma} \sum_k \delta_k P_{jkt}^{1-\sigma} C_{jt}^{(1-\sigma)(\varepsilon_k-1)} = E_{jt}^{1-\sigma} \quad (\text{A.17})$$

Define the aggregate price index  $P_{jt}$  by  $P_{jt} C_{jt} = \sum_k P_{jkt} C_{jkt}$ . Equation (A.17) then implies:

$$P_{jt} = \left[ \sum_{n=a,m,s} \delta_n P_{jnt}^{1-\sigma} C_{jt}^{(1-\sigma)(\varepsilon_n-1)} \right]^{\frac{1}{1-\sigma}} \quad (\text{A.18})$$

The aggregate price index is therefore the expenditure-deflating price index associated with the household's non-homothetic CES aggregator. Equation (A.13) can consequently be written as:

$$C_{jkt} = \delta_k \left( \frac{P_{jkt}}{P_{jt}} \right)^{-\sigma} C_{jt}^{\varepsilon_k(1-\sigma)+\sigma} \quad \forall k = a, m, s \quad (\text{A.19})$$

The expenditure share of sector  $k$  is:

$$\begin{aligned} e_{jkt} &= \frac{P_{jkt} C_{jkt}}{P_{jt} C_{jt}} \\ &= \delta_k \left( \frac{P_{jkt}}{P_{jt}} \right)^{1-\sigma} C_{jt}^{(1-\sigma)(\varepsilon_k-1)} \quad \forall k \end{aligned} \quad (\text{A.20})$$

Relative to manufacturing:

$$\begin{aligned} \Rightarrow \frac{e_{jkt}}{e_{jmt}} &= \frac{\delta_k}{\delta_m} \left( \frac{P_{jkt}}{P_{jmt}} \right)^{1-\sigma} \times \frac{C_{jt}^{(\varepsilon_k-1)(1-\sigma)}}{C_{jt}^{(\varepsilon_m-1)(1-\sigma)}} \\ &= \frac{\delta_k}{\delta_m} \left( \frac{P_{jkt}}{P_{jmt}} \right)^{1-\sigma} C_{jt}^{(\varepsilon_k-\varepsilon_m)(1-\sigma)} \end{aligned} \quad (\text{A.21})$$

Equation (A.20) can also be written in terms of total expenditure:

$$\begin{aligned} e_{jkt} &= \delta_k \left( \frac{P_{jkt}}{E_{jt} C_{jt}^{-1}} \right)^{1-\sigma} C_{jt}^{(1-\sigma)(\varepsilon_k-1)} \\ &= \delta_k \left( \frac{P_{jkt}}{E_{jt}} \right)^{1-\sigma} C_{jt}^{(1-\sigma)\varepsilon_k} \quad \forall k = a, m, s \end{aligned} \quad (\text{A.22})$$

Hence,

$$C_{jt} = \left[ \left( \frac{e_{jkt}}{\delta_k} \right)^{\frac{1}{1-\sigma}} \left( \frac{E_{jt}}{P_{jkt}} \right) \right]^{1/\varepsilon_k} \quad k = a, m, s \quad (\text{A.23})$$

Substituting this expression into Equation (A.21) gives:

$$\begin{aligned} \frac{e_{jkt}}{e_{jmt}} &= \frac{\delta_k}{\delta_m} \left( \frac{P_{jkt}}{P_{jmt}} \right)^{1-\sigma} \left[ \frac{E_{jt}}{P_{jmt}} \left( \frac{e_{jmt}}{\delta_m} \right)^{\frac{1}{1-\sigma}} \right]^{\frac{(1-\sigma)(\varepsilon_k-\varepsilon_m)}{\varepsilon_m}} \\ &= \frac{\delta_k}{\delta_m} \left( \frac{P_{jkt}}{P_{jmt}} \right)^{1-\sigma} \left( \frac{e_{jmt}}{\delta_m} \right)^{\frac{\varepsilon_k}{\varepsilon_m}-1} \left( \frac{E_{jt}}{P_{jmt}} \right)^{\frac{(1-\sigma)(\varepsilon_k-\varepsilon_m)}{\varepsilon_m}} \end{aligned} \quad (\text{A.24})$$

Thus, sector  $k$ 's expenditure share can be expressed as:

$$e_{jkt} = \delta_k \left( \frac{e_{jmt}}{\delta_m} \right)^{\frac{\varepsilon_k}{\varepsilon_m}} \left( \frac{P_{jkt}}{P_{jmt}} \right)^{1-\sigma} \left( \frac{E_{jt}}{P_{jmt}} \right)^{(1-\sigma)\left(\frac{\varepsilon_k}{\varepsilon_m}-1\right)} \quad (\text{A.25})$$

This final expression relates sector  $k$ 's expenditure share to the manufacturing expenditure share, relative prices, and real aggregate expenditure. It provides the operational form of the household's demand system used to connect the model to the expenditure-share evidence in Section 2.

## A.5 Existence of a Static Equilibrium

Fix the technology state  $\{A_{jkt}(v), A_{ikt}(v)\}_{v \in [0,1]}$  and the leadership sets  $\{\Theta_{jkt}, \Theta_{ikt}\}_{k=a,m,s}$ . For each country  $j \in \{H, F\}$ , normalize prices by the domestic wage,  $\hat{P}_{jkt} := P_{jkt}/w_{jt}$ . Given the fixed technology state and leadership sets, the within-period allocation is characterized by the final-good producers' demands, the monopolists' pricing decisions, and the household demand system.

Under the regularity conditions imposed in the model, the resulting excess-demand correspondence is continuous and homogeneous of degree zero in prices. The homogeneity follows from the fact that demands, profits, and the household allocation depend on prices through relative prices and the corresponding income effects. Standard boundary conditions ensure that normalized excess demand does

not point outward at the boundary of the price simplex. The Debreu–Gale–Nikaido existence theorem therefore implies the existence of a normalized price vector at which all markets clear, with the final market clearing by Walras’ law.

The result establishes existence of a within-period equilibrium conditional on the technology state and leadership sets. It does not characterize the stochastic evolution of those objects across periods. In particular, a dynamic equilibrium would require specifying the joint evolution of innovation outcomes and the resulting leadership sets, which is beyond the scope of the present analysis.

## A.6 Innovation Race

This appendix derives the innovation incentives for a single intermediate variety  $v$  in sector  $k$  when two countries compete for technological leadership. Suppose country  $j$  is the current leader, so that  $A_{jkt}(v) = \hat{A}_{kt}(v)$ , while country  $i$  is the follower. Define the follower’s relative productivity as

$$g_{jkt}(v) := \frac{A_{ikt}(v)}{A_{jkt}(v)} \in (0, 1].$$

Both countries follow the transition rule in Equation (4.6), under which a successful innovation multiplies current productivity by  $\gamma_{jk}$ .

**The overtake condition.** If the leader fails and the follower succeeds, the leader remains at  $\hat{A}_{kt}(v)$  while the follower reaches  $\gamma_{jk}g_{jkt}(v)\hat{A}_{kt}(v)$ . The follower therefore overtakes if and only if

$$\gamma_{jk}g_{jkt}(v) > 1 \iff g_{jkt}(v) > \frac{1}{\gamma_{jk}}. \quad (\text{A.26})$$

This defines the close case. When  $g_{jkt}(v) \leq 1/\gamma_{jk}$ , the variety is in the far case. In the far case, a successful innovation by the follower is insufficient to overtake the incumbent, so the follower’s current innovation effort does not affect the leader’s continuation payoff.

**The leader’s innovation incentive.** Define the market-size term

$$B_{jkt+1} := L_{jkt+1} + \left( \frac{P_{ikt+1}}{\tau_{jik}P_{jkt+1}} \right)^{\frac{1}{1-\alpha}} L_{ikt+1}. \quad (\text{A.27})$$

Using Equation (4.4), the leader’s expected continuation payoff, net of R&D expenditure, is:

$$\begin{aligned} V^{\text{leader}} = & \mu_{jkt}(v) \cdot \beta \pi P_{jkt+1} B_{jkt+1} \gamma_{jk} \hat{A}_{kt}(v) \\ & + (1 - \mu_{jkt}(v)) (1 - \mu_{ikt}(v)) \cdot \beta \pi P_{jkt+1} B_{jkt+1} \hat{A}_{kt}(v) \\ & + (1 - \mu_{jkt}(v)) \mu_{ikt}(v) \cdot \mathbf{1}_{\{g_{jkt}(v) > 1/\gamma_{jk}\}} \cdot 0 \\ & - \lambda_j^{-1} P_{jkt} F(\mu_{jkt}(v)) \gamma_{jk} \hat{A}_{kt}(v). \end{aligned} \quad (\text{A.28})$$

In the close case, failure by the leader combined with success by the follower results in the loss of the variety. In the far case, the same event leaves the leader in control. The corresponding continuation

payoffs are therefore:

$$V^{\text{leader, close}} / \hat{A}_{kt}(\mathbf{v}) = \beta \pi P_{jkt+1} B_{jkt+1} \left[ \gamma_{jk} \mu_{jkt}(\mathbf{v}) + (1 - \mu_{jkt}(\mathbf{v})) (1 - \mu_{ikt}(\mathbf{v})) \right] - \lambda_j^{-1} P_{jkt} F(\mu_{jkt}(\mathbf{v})) \gamma_{jk}, \quad (\text{A.29})$$

$$V^{\text{leader, far}} / \hat{A}_{kt}(\mathbf{v}) = \beta \pi P_{jkt+1} B_{jkt+1} \left[ 1 + \mu_{jkt}(\mathbf{v}) (\gamma_{jk} - 1) \right] - \lambda_j^{-1} P_{jkt} F(\mu_{jkt}(\mathbf{v})) \gamma_{jk}. \quad (\text{A.30})$$

The first-order conditions are:

$$F'(\mu_{jkt}(\mathbf{v})) = \lambda_j \beta \pi \frac{P_{jkt+1}}{P_{jkt}} B_{jkt+1} \frac{\gamma_{jk} - 1 + \mu_{ikt}(\mathbf{v})}{\gamma_{jk}} \quad (\text{close}), \quad (\text{A.31})$$

$$F'(\mu_{jkt}(\mathbf{v})) = \lambda_j \beta \pi \frac{P_{jkt+1}}{P_{jkt}} B_{jkt+1} \frac{\gamma_{jk} - 1}{\gamma_{jk}} \quad (\text{far}). \quad (\text{A.32})$$

When  $F(\mu) = \frac{1}{2} \mu^2$ , these conditions reduce directly to Equations (4.7) and (4.8). The leader's innovation incentive therefore depends on the follower's effort only in the close case, when the follower's innovation can displace the incumbent.

**The follower's innovation incentive.** The follower's expected payoff is:

$$V^{\text{follower}} = \mu_{ikt}(\mathbf{v}) (1 - \mu_{jkt}(\mathbf{v})) \cdot \mathbf{1}_{\{g_{jkt}(\mathbf{v}) > 1/\gamma_{jk}\}} \cdot \beta \pi P_{ikt+1} B_{ikt+1} \gamma_{jk} g_{jkt}(\mathbf{v}) \hat{A}_{kt}(\mathbf{v}) - \lambda_i^{-1} P_{ikt} F(\mu_{ikt}(\mathbf{v})) \gamma_{jk} g_{jkt}(\mathbf{v}) \hat{A}_{kt}(\mathbf{v}), \quad (\text{A.33})$$

where

$$B_{ikt+1} := L_{ikt+1} + \left( \frac{P_{jkt+1}}{\tau_{ijk} P_{ikt+1}} \right)^{\frac{1}{1-\alpha}} L_{jkt+1}.$$

The follower earns a positive continuation payoff only when it succeeds, the leader fails, and the initial productivity gap is sufficiently small to be bridged by one innovation. In the close case, the first-order condition is:

$$F'(\mu_{ikt}(\mathbf{v})) = \lambda_i \beta \pi \frac{P_{ikt+1}}{P_{ikt}} B_{ikt+1} (1 - \mu_{jkt}(\mathbf{v})) \quad (\text{close}), \quad (\text{A.34})$$

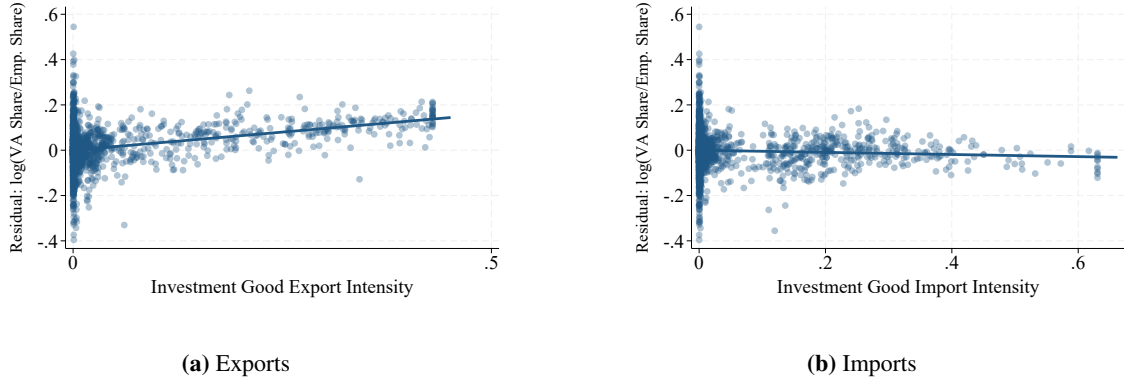
which yields Equation (4.9) when  $F(\mu) = \frac{1}{2} \mu^2$ .

In the far case, the follower cannot overtake even after a successful innovation. The payoff-relevant indicator in Equation (A.33) is therefore zero, leaving only the R&D cost. Since  $F(0) = 0$  and  $F' > 0$ , the optimal choice is  $\mu_{ikt}(\mathbf{v}) = 0$ .

The innovation race is thus asymmetric despite the symmetric transition rule. A leader earns the continuation value of the variety whenever it retains technological leadership, whereas a follower earns that value only upon successful replacement of the incumbent. Consequently, the follower's innovation incentive is positive only when the productivity gap can be closed in a single step, while the leader's incentive reflects both the gain from further innovation and the risk of replacement in the close case.

## B Output Appendix

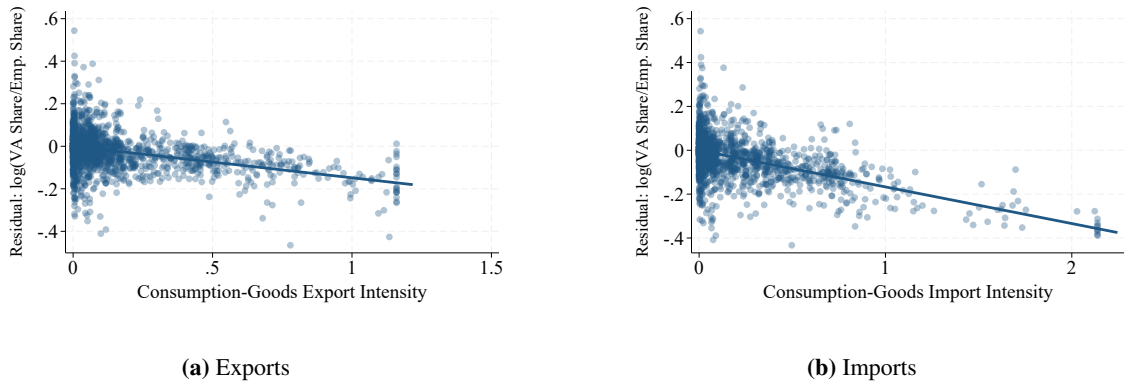
This appendix reports the partial-residual plots for investment-good and final-consumption-good trade referenced in Section 2, complementing the intermediate-good results in Figures 2 and 3 in the main text. Figures B.1 and B.2 report the value-added–employment comparison for these two trade margins, while Figures B.3 and B.4 report the corresponding relationship with the profit share of value added.



**Figure B.1:** Investment-Good Trade and the Log Ratio

*Notes:* This figure shows investment-good export intensity (panel a) and investment-good import intensity (panel b).

Figure B.1 plots the same orthogonalized log value-added–employment ratio used in Figure 2, replacing intermediate-good trade intensity with investment-good trade intensity. Investment-good exports are positively related to the ratio, although the support of the export measure is substantially narrower than that of intermediate-good exports, with investment-good export intensity rarely exceeding 0.5 compared with values above 2.5 for intermediate goods. The relationship with investment-good imports is comparatively weak and close to flat. Figure B.2 shows a different pattern for final-consumption-good trade: exports are negatively related to the ratio, whereas imports exhibit a pronounced negative relationship, similar in direction and visual magnitude to the intermediate-good import relationship reported in the main text.

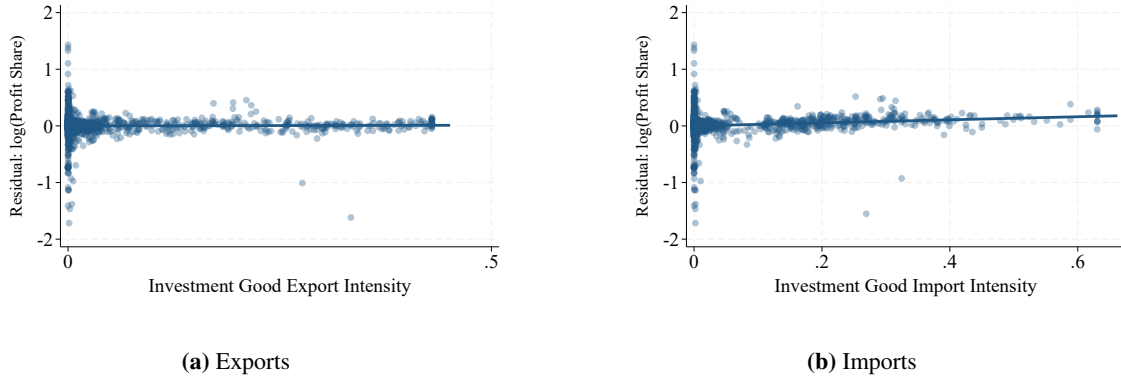


**Figure B.2:** Final-Consumption-Good Trade and the Log Ratio

*Notes:* This figure shows final-consumption-good export intensity (panel a) and final-consumption-good import intensity (panel b).

The next two figures examine the profit share of value added, the outcome variable in Table 2.

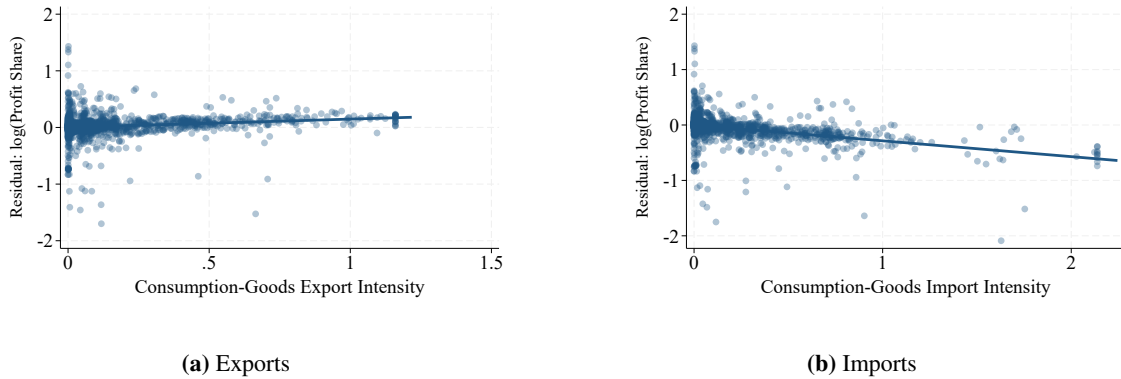
Figure B.3 plots the profit share against investment-good export and import intensity using the same partial-residual approach as Figure 3 in the main text.



**Figure B.3:** Investment-Good Trade and the Profit Share

*Notes:* This figure shows investment-good export intensity (panel a) and investment-good import intensity (panel b).

Investment-good exports show little relationship with the profit share, whereas investment-good imports exhibit a mild positive slope. The latter differs in sign from the negative relationship between intermediate-good imports and the profit share shown in the main text. Figure B.4 completes the comparison for final-consumption-good trade. Final-consumption-good exports show a mild positive relationship with the profit share, while imports exhibit a pronounced negative relationship. Thus, among the trade margins considered in this appendix, final-consumption-good imports are the only margin that displays the same sign for both the value-added–employment ratio and the profit share as the corresponding intermediate-good import relationship in the main text.



**Figure B.4:** Final-Consumption-Good Trade and the Profit Share

*Notes:* This figure shows final-consumption-good export intensity (panel a) and final-consumption-good import intensity (panel b).